No False Green: How Output Schemas and Proxy Columns Lead Language Models to Report Unevidenced Compliance Obligations as Satisfied

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Abstract. A compliance dashboard that marks an obligation satisfied when the evidence to assess it was never present makes the one error its user cannot see: a **false green**. We study when language models make it in employer-side US immigration compliance, where a system reads an employer's exports and returns `pass`, `fail`, `needs_evidence` or `unmonitored` per obligation and entity. A generator fixes every label by construction and can add **proxy columns**, fields that sit beside the required record without being it, such as an onboarding date beside a missing Form I-9. In a preregistered study of seven models from five developers, proxy columns raised the false-green rate by 0.427 under a pass/fail dashboard schema and by 0.217 under a schema that offers abstention; on proxy-column firms, removing the non-committal verdicts raised it by 0.259 and redefining both verdicts by 0.084 (exact permutation $p \leq 0.0013$, Holm). The effects survive a reworded prompt, replicate on ten further firms, and are carried mostly by date columns that models cite as their evidence. An exploratory study of thirteen models on a hand-authored firm generated these hypotheses. No attorney validated the labels; all data, prompts and replies are released.

1 Introduction

A green cell on a compliance dashboard asserts that an obligation, attached to a worker or a filing, was assessed and found satisfied. It silently assumes the evidence was present. When it was not, the system can say so or render the cell green anyway, and a mostly green dashboard looks like a working product. We call the second choice a **false green**.

A false green is not the mirror image of a false alarm. A false alarm costs review time and is corrected by the review it triggers; a false green triggers nothing, and the review it suppresses is the only thing that would surface it. In immigration compliance the cost is concrete: Form I-9 paperwork violations are penalised per form (8 C.F.R. 274a.10(b)(2)), and violations of a Labor Condition Application (LCA), the wage attestation behind an H-1B petition, per violation (20 C.F.R. 655.810(b)). A gap found before an inspection can often be cured; found during one, it becomes a finding.

Pass/fail evaluation cannot see it. An evaluation with no category for "the evidence was not there" never penalises a system that answers `pass` whenever evidence is absent. Recent benchmarks score overclaiming under insufficient evidence for agents (Solozobov, 2026; Tallam, 2026), and legal and tabular benchmarks include a "not enough information" answer (§2); none scores regulatory verdicts over a third party's operational records when the required record is missing, where declining to decide is the correct answer for about half the items.

We built an instrument for this error (Figure 1) and report three findings. **Proxy columns are the largest cause we found:** planting columns that resemble required evidence, with the prompt unchanged, raises the false-green rate by 0.217 to 0.427, and models cite those columns as their evidence. **The output schema matters in two parts:** removing the verdicts for "could not tell" and redefining `pass` from "shown to be satisfied" to "satisfied" each raise false greens, more on firms with proxy columns. **High recall does not bring safety:** no model that detects nearly every violation comes close to zero false greens.

Our contributions are: a four-verdict task with a stated evidentiary standard (§3); an instrument whose generator fixes labels before data, renders paired firms with and without proxy columns, and whose labels are re-derived by a separately written checker (§4); a preregistered study with two declared follow-ups (§6) and the exploratory study that motivated it (§7); and a release in which every number in the paper is recomputed from retained replies by a script.

2 Related work

Legal NLP and compliance. Legal benchmarks (Guha et al., 2023; Chalkidis et al., 2022; Zheng et al., 2021; Hendrycks et al., 2021; Fei et al., 2024) evaluate reasoning over legal text, rule application (Zhou et al., 2025; Dou et al., 2026) and unsupported statements about the law (Dahl et al., 2024; Magesh et al., 2025). Immigration datasets are petitioner- and text-side (Shportun, 2026; Afsharrad and Mousavi, 2026). Compliance checking predates language models (Zhang and El-Gohary, 2016; Amaral et al., 2022); recent LLM work makes the model the regulated subject (Guldimann et al., 2024) or the assessor of another party's documentation (Marino et al., 2025; Yang et al., 2026; Wang et al., 2025; Sarkar, 2026; Luo et al., 2026). Absent support is already a correct answer over legal text (ContractNLI's Not-Mentioned, Koreeda and Manning, 2021; unanswerable privacy questions, Ravichander et al., 2019); what is new here is a correct answer for a missing *primary record* when obligations are assessed from a third party's operational exports.

Overclaiming under insufficient evidence. DEMM-Bench labels by construction whether agent records suffice and reports overclaims, including a "container fallacy" in which a reassuring artefact stands in for evidence (Solozobov, 2026); Partial Evidence Bench scores unsafe completeness claims, including in compliance audits, when evidence lies outside an agent's permissions (Tallam, 2026). We do not claim the directional, construction-labelled framing as new;

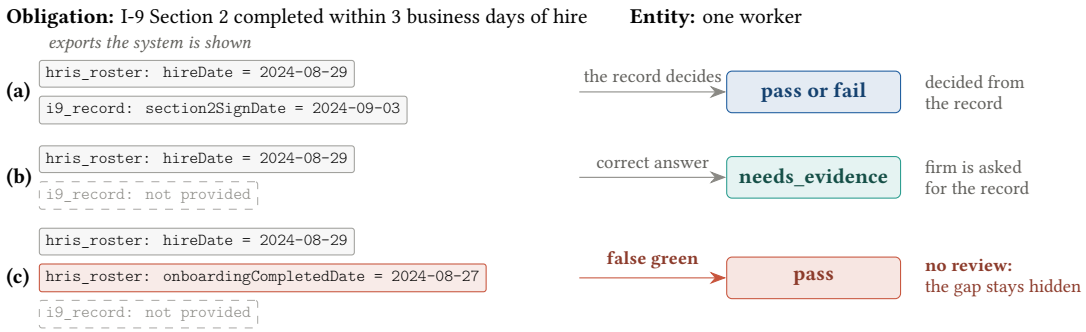


Figure 1. The task and the error. Each panel shows the exports a system is given for one obligation and one worker (values illustrative). With the I-9 record present the record decides (a); with it missing the correct verdict is needs_evidence (b); with it missing but an onboarding date beside it, a model may answer pass (c). A wrong fail sends the item to review; a wrong pass removes it from review.

we differ in subject, in the source of insufficiency (a missing export rather than permissions), and in manipulating proxy columns and verdict menus directly across models.

Insufficiency, distractors and answer menus. Selective prediction treats abstention as a cost (Chow, 1970; El-Yaniv and Wiener, 2010; Geifman and El-Yaniv, 2017); language-model abstention is studied mostly on questions (Rajpurkar et al., 2018; Yin et al., 2023; Kirichenko et al., 2025; Wen et al., 2025; Kadavath et al., 2022), and models answer rather than abstain when context is insufficient (Joren et al., 2024). NotEnoughInfo or unknown labels exist for text and tables (Thorne et al., 2018; Aly et al., 2021; Wang et al., 2021; Theologitis et al., 2026) and unanswerable database questions (Lee et al., 2022; Wang et al., 2023; Lee et al., 2024). Irrelevant content changes answers whose label it leaves fixed (Shi et al., 2023), including irrelevant tax forms in rule application (Zhou et al., 2025), an instance of shortcut reliance (Geirhos et al., 2020); our proxy columns are the case where the distractor resembles the missing evidence and the error has a direction. Answer menus change behaviour: expanding a binary label set with "insufficient evidence" makes models more conservative (Fu et al., 2026), the presence and framing of an abstention option change how often models commit wrongly (Guo et al., 2026; Madhusudhan et al., 2025). In CLEAR, adding "I don't know" to multiple-choice medical questions *increased* wrong answers (Guo et al., 2026); here the abstaining verdicts are defined by whether the required record is present, and offering them lowers false greens (§6.2), so the effect of an abstention option depends on how it is defined, and output formats and forced choice alter answers (Tam et al., 2024; Röttger et al., 2024; Sclar et al., 2024), while grading that gives no credit for uncertainty rewards guessing (Kalai et al., 2025). Our schema conditions place these effects in a compliance interface and separate removing the abstaining verdicts from redefining the committal ones.

3 Task and evidence standard

An instance is a firm with source exports, entities (workers, LCAs, the employer), a rulebook of obligations each citing a regulation and listing the information needed to decide it, and a pinned evaluation date. A **pair** is an obligation and an entity it applies to; the target is one of four verdicts (Table 1). Columns are matched to the required information by meaning, and reference knowledge such as holidays is the system's to supply. If a required item is absent for the entity, the pair is **unassessable** and any verdict but `needs_evidence` is wrong, whether or not the underlying situation complies.

Table 1. The four verdicts.

Verdict	Meaning
`pass`	Required evidence present; obligation met
`fail`	Required evidence present; obligation breached
`needs_evidence`	A required record is absent
`unmonitored`	No employer export could evidence it

What counts as evidence. We score a primary-record standard: an obligation is evidenced only by the record the law requires the employer to keep. For I-9 timeliness that is the form, whose dates the employer must retain and produce for inspection (8 C.F.R. 274a.2(b)(2)); for the LCA wage it is the payroll record of wages paid (20 C.F.R. 655.731(b)(1), (c)). An HR status, onboarding date or salary of record is the employer's statement about those records. A monitoring product's green cell implies an inspector would find the record in order, which only the record supports. A proxy-accepting standard is defensible for triage, and a reader who prefers it should read our proxy results as a proxy-acceptance rate.

Metrics. The **false-green rate** (FGR) is the share of unassessable pairs answered `pass`. Answering `fail` on such a pair is also wrong, but it triggers review, so it is not counted. A pair with no parsed answer can never be a false green, so every table reports coverage. FGR alone is gameable: `always_fail` scores 0 and catches every violation. Violation recall and the false-alarm rate on conforming controls (any verdict but `pass`) must therefore be reported with it (Appendix D gives trivial baselines). `unmonitored` on an unassessable pair is wrong but not a false green; in practice it acts as a second abstention (Appendix Figure 8).

4 The instrument

Rulebook. 194 obligations across ten US employment-immigration regimes, each with its check, threshold, citation and required information, built from the registry of a commercial compliance engine developed by the author's company, whose citations its maintainers had checked against primary sources.

Generator. Labels precede data: for each pair the generator draws a disposition (a planted violation, a conforming control with a share placed exactly on the legal boundary, or withheld evidence) and then writes records that decide it. It renders four rule families completely: I-9 Section 1 and Section 2 timeliness, reverification before authorisation expires, and the LCA required wage. Evidence is withheld a whole export at a time, since a blank date for a past deadline would read as "not done", a violation. A firm has twelve workers and 42 pairs.

Proxy columns. For Study 3 the generator can add five roster columns that sit beside a rule's evidence without being it: an I-9 status of "Complete", an onboarding date, the current authorisation's end date, a salary of record set at the LCA offered wage, and a pay-band midpoint. They appear whether or not the export they imitate is withheld, and come from a separate random stream, so a plain firm and its proxy twin have identical labels and other exports (Appendix Figure 7). The wage rule has a native proxy even in plain firms: the LCA's offered wage beside a missing actual-wage table.

Label validation. A separate AI agent wrote a deterministic checker from the regulations without access to the generator or any label. It reproduces all 1,176 distinct generated labels after exposing and fixing one generator bug, a year-end observed holiday that touched no study firm. Both programs were written by agents of one model family and share the primary-record standard; boundary choices are stated with their sensitivity (Appendix G).

Anchor firm. A hand-authored employer of twelve workers with eleven exports and 221 pairs, whose labels were the engine's acceptance gate, where `fail` means "requires action". It is the weakest component. Two blind AI audits of its 120 evaluated evidence requests overturned 2 and 5 labels with none in common (Cohen's $\kappa = -0.024$), and the second auditor found only 7 of its 20 violation labels shown by the exports. We use it only as a realism probe for false greens, which do not depend on violation labels (Appendix D).

5 Protocol

Each request carries the relevant exports verbatim, the obligations with their required information, and twelve pairs; the reply gives a verdict and the evidence field relied on per pair. The prompt is frozen (version 1.1.0, Appendix B); only the verdict menu changes across four conditions (Table 2).

The step from *offered* to *closed* is the **abstention effect**; from *closed* to *dashboard*, the **definition-and-warning effect**, which cannot separate its two changes. Thirteen models from eight developers were run with temperature 0 and a 32,000-token cap (Appendix Table 6): six free-tier models, and seven on Google's Vertex AI paid from cloud credits. Unparseable lines are never coerced to a verdict (we report the unparseable rate), delivery failures are re-requested, and the commercial engine is not a baseline, since it and the anchor labels were developed together (Appendix C).

6 A preregistered study of schemas and proxy columns

6.1 Design

Study 3 was declared in a file committed before any call (commit `e1baaeb`), with its analysis script: twelve fresh generated firms (504 pairs, 180 unassessable), each rendered plain and with proxy columns, under the four schemas, with seven models from five developers, once per cell; every cell answered at least 456 of 504 pairs. Contrasts are paired on pairs answered in both cells and averaged over models with equal weight. The declared inference is a bootstrap over the 18 (firm, rule) clusters of unassessable pairs. Four hypotheses predicted increases: **P1** the abstention effect and **P2** the definition-and-warning effect, on proxy-column firms; **P3** and **P4** the proxy-column effect under the dashboard and offered schemas.

6.2 Results

All four were confirmed (Table 3, Figure 2). The bootstrap p-values all sit at its floor (0.002 after correction), so we add an exact sign-flip permutation test, which gives Holm-adjusted $p \le 0.0013$ for all four. Two features of the design temper the headline numbers. The 18 clusters are unbalanced: the I-9 pairs (99) come from only three firms that withhold the I-9 export, and the wage pairs (81) from nine. Weighting the twelve firms equally instead of the pairs, every effect stays positive and significant (exact p from < 0.001 to 0.028), but P3 falls to +0.244, because the large I-9 effect then counts for three firms rather than 99 pairs. Dropping any one model or any one firm leaves every effect positive.

The models differ (Figure 3). The proxy-column effect under the dashboard schema is significant for all seven, and under the offered schema for five (not Gemini 3.1 Pro or Kimi K2). The abstention effect is significant for five (not Gemma 4 or gpt-oss-120b), the definition-and-warning effect for one (gpt-oss-120b). Removing only `needs_evidence` raised false greens by 0.080 pooled ($p = 0.004$). The declared interaction, a larger definition-and-warning effect on proxy-column firms, was not found (+0.048 [−0.01, 0.10], $p = 0.10$). On plain firms both schema effects are present but smaller (exploratory: +0.093 and +0.035).

6.3 Two declared follow-ups

Study 3b: a reworded prompt. Each condition had been tested in one wording. We reworded every instruction sentence of the offered and dashboard prompts, keeping mean-

Table 2. The schema conditions.

Condition	Verdicts	Change from the row above
offered	all four	—
removed-only	no `needs_evidence`	Abstention verdict and three sentences about it removed
closed	`pass`, `fail`	`unmonitored` removed; every pair must be decided
dashboard	`pass`, `fail`	Both verdicts redefined as "the obligation is satisfied / violated"; warning sentence removed

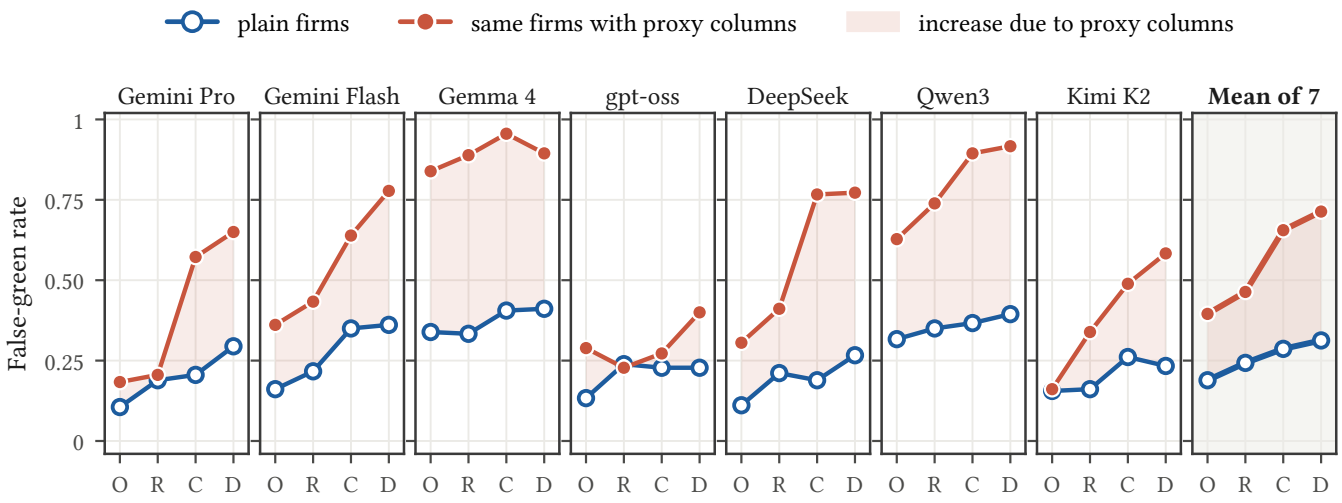


Figure 2. False-green rate on the 180 unassessable pairs of the Study 3 firms under the four schemas (O offered, R removed-only, C closed, D dashboard), for plain firms and the same firms with proxy columns. Shading marks where proxy columns raise the rate. Rates are per cell; the tests in Table 3 are paired.

ing and verdict tokens, and re-ran the Study 3 firms with the three Google models (declared before any call; the four other developers' models were not re-run, because their cloud billing had not yet posted). All three declared tests replicated (Holm-adjusted p = 0.0015 each, the bootstrap's floor). Proxy columns raised the false-green rate by 0.483 under the reworded dashboard schema and by 0.301 under the reworded offered schema, against 0.440 and 0.265 for the same three models in the original wording; moving from the offered to the dashboard schema on proxy-column firms raised it by 0.340, against 0.321.

Study 3c: more I-9 firms, one column at a time. To test the I-9 effect on more than three firms, and to ask which column carries it, we drew ten fresh firms that all withhold the I-9 export and rendered each plain, with all five proxy columns, and with each I-9 proxy column alone, with the same three models (declared before any call; firm-level exact permutation tests; Figure 4). In Study 3 the I-9 effect rested on three firms: models called 0.000 to 0.023 of plain-firm I-9 pairs compliant across the four schemas, and 0.193 under the offered and 0.686 under the dashboard schema once proxy columns were present (the wage rule, whose plain firms already show the LCA's offered wage, drew false greens on 0.442 of pairs even there with abstention offered). On the ten new firms all five declared tests were confirmed (exact $p \le 0.004$, Holm-adjusted ≤ 0.010, positive in at least nine firms). All five columns together raised the I-9 false-green rate by 0.822 under the dashboard schema and by 0.302 under the offered schema. One column at a time, the onboarding date raised it by 0.509 on Section 1 and 2 pairs and the current authorisation's end date by 0.759 on reverification pairs, while the I-9 status flag raised it by only 0.097, almost entirely through Gemma 4. Models take a date or an amount that looks like the required one as evidence far more readily than a status flag. The reverification case is the most defensible proxy: a current authorisation implies, without showing, a timely reverification.

6.4 What models rely on

Every reply line names the evidence field relied on, so the mechanism can be read directly, with the caveat that this is the model's own report and stated reasons need not be the cause (Turpin et al., 2023); the causal evidence is the paired contrast and Study 3c (Table 4). On proxy-column firms, 1,376 of the 2,807 false greens (49%) cite one of the five planted columns, and for the I-9 rules under the offered schema 118 of 134 (88%), almost all the onboarding date or the current authorisation's end date. On plain firms, wage-rule false greens most often cite the LCA's offered wage. Under the dashboard schema more false greens cite other fields, consistent with a schema that asks only whether the obligation is satisfied.

Detection does not bring safety. On the proxy-column firms, Gemini 3.1 Pro, Gemini 3.8 Flash, gpt-oss-120b and

Table 3. The four declared tests of Study 3, with robustness checks.

Test	Effect [95% CI]	Exact p (Holm)	Firm-weighted effect, exact p	Drop one model	Drop one firm
P1 Abstention (proxy firms)	+0.259 [+0.20, +0.33]	<0.001	+0.203 (p < 0.001)	+0.226 to +0.306	+0.227 to +0.271
P2 Definition and warning (proxy firms)	+0.084 [+0.04, +0.12]	0.001	+0.051 (p = 0.028)	+0.067 to +0.099	+0.065 to +0.091
P3 Proxy columns, dashboard	+0.427 [+0.29, +0.56]	<0.001	+0.244 (p = 0.021)	+0.407 to +0.469	+0.353 to +0.449
P4 Proxy columns, offered	+0.217 [+0.15, +0.29]	<0.001	+0.237 (p = 0.002)	+0.170 to +0.244	+0.200 to +0.230

Effect: change in false-green rate, pooled over seven models with equal weight, paired on unassessable pairs answered in both cells; interval from the declared bootstrap over 18 firm-by-rule clusters. Exact p: sign-flip permutation over those 18 clusters (all 262,144 patterns), Holm across the four tests. Firm-weighted: each of the 12 firms weighted equally, exact sign-flip over firms (4,096 patterns). Drop one: range of the pooled effect with any one model, or any one firm, left out. All columns after the first were computed after the results.

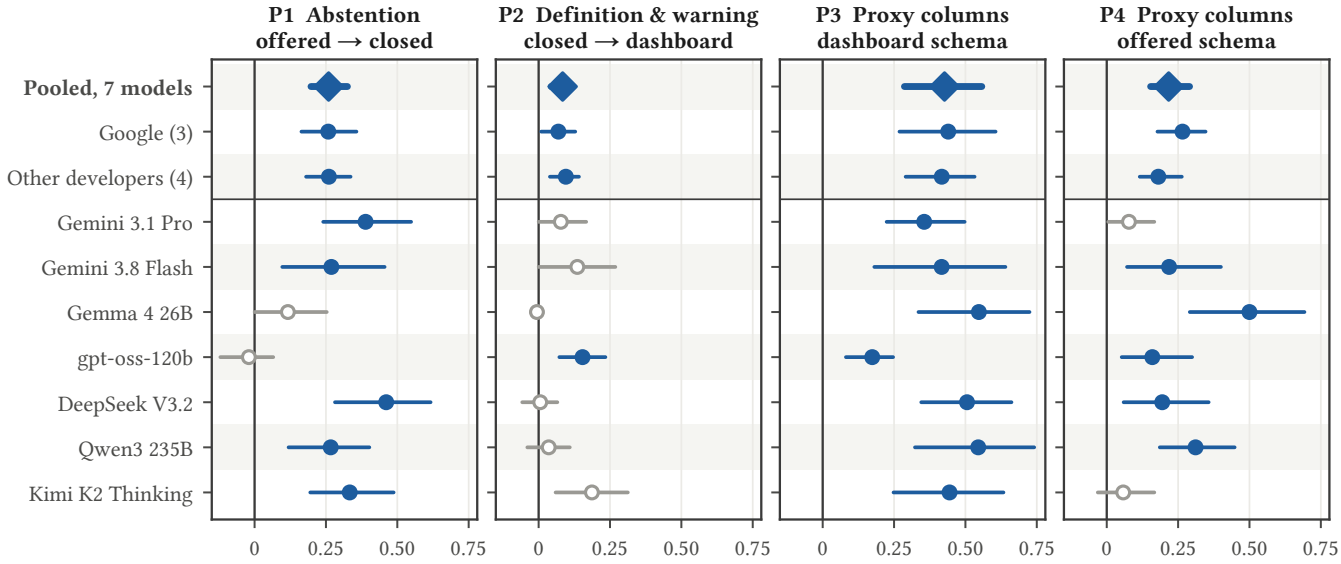


Figure 3. The declared tests per model and by developer group, with 95% bootstrap intervals. Filled markers are significant after Holm correction within their family; hollow grey markers are not. Pooled effects are robust; per-model effects vary, most of all P2.

Kimi K2 detect at least 0.920 of violations with abstention offered, yet call 0.161 to 0.361 of unassessable pairs compliant. Across the seven models recall and false-green rate are negatively related, so higher recall helps, but none of the high-recall models comes near zero (Appendix Figure 9). Proxy columns change the false-alarm rate on conforming controls by at most 0.038 with abstention offered and 0.080 under any schema, so their cost falls mostly on the false-green side.

7 The exploratory study that generated the hypotheses

Before Study 3, all thirteen models were run on a declared 140-pair subset of the anchor firm, not all 221 pairs, for the free-tier budget (every planted violation and 120 evidence requests), most once, three Google models three times (their runs differed in false-green rate by at most 0.142).

With abstention offered the thirteen models report 0.042 to 0.153 of unassessable pairs as satisfied, and 0.131 to 0.589 under the dashboard schema (Figure 5); the dashboard schema raised the rate significantly for eight of the nine models tested for it, and the four added later were run descriptively. Decomposing the schema for the three repeated-run models showed that each failed through a different route: Gemma 4 when `needs_evidence` alone was removed, Gemini 3.8 Flash when every non-committal verdict was removed, and Gemini 3.1 Pro only when `pass` was redefined; Pro otherwise sent unassessable pairs to `unmonitored` or `fail`. False greens concentrated where a populated column sat beside the missing evidence. Two of the excerpts selected by a rule fixed before any reply was read: Qwen 3.8 27B and dots.3 cited an unitemized payroll deduction total as evidence that no prohibited fee was charged, and Nemotron 3.5 Lightning and Laguna XS 2.1 inferred the physical examination of I-9 documents from the presence of a document title. These observations are what Study 3 tested.

8 Discussion

Three design rules follow. Keep a verdict for "could not tell"; define `pass` as "the required evidence is present and satisfies the obligation"; and treat any export column that resembles required evidence as a hazard, by withholding it from the model or routing the obligations it touches to review, since no schema tested protected against it. With the first two in place false-green rates of 0.04 to 0.15 remain on the anchor firm and 0.16 to 0.84 on proxy-column firms. That models respond to wording is well known (Sclar et al., 2024); the

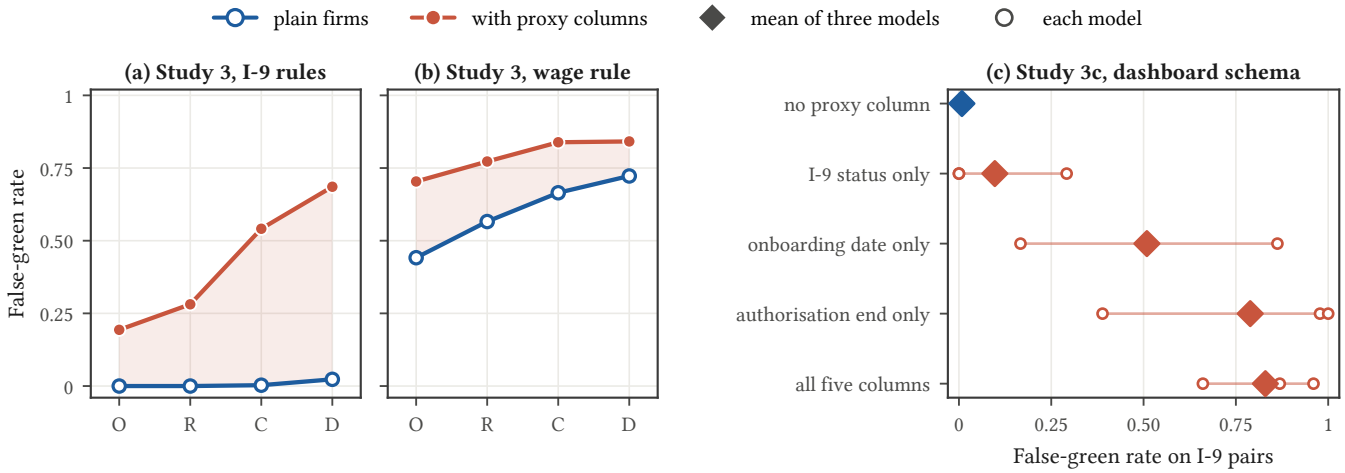


Figure 4. Where proxy columns act. (a, b) Study 3 false-green rates by rule family under the four schemas (I-9 pairs from 3 firms, wage pairs from 9). (c) Study 3c, ten further firms that withhold the I-9 export, dashboard schema: with no proxy column, with each I-9 proxy column alone (scored on the pairs it bears on) and with all five; diamonds are the mean of three models, circles the models.

Table 4. The column cited as evidence on each false green, proxy-column firms.

Rule family	Schema	False greens	Citing a planted proxy column	Per model	Citing the LCA offered wage
I-9 (three rules)	offered	134	118 (88%)	50%–100%	0 (0%)
I-9 (three rules)	dashboard	462	276 (60%)	0%–100%	0 (0%)
LCA required wage	offered	364	123 (34%)	4%–77%	119 (33%)
LCA required wage	dashboard	437	124 (28%)	0%–71%	45 (10%)

Study 3 proxy-column firms, pooled over the seven models; per model: range of the share citing a planted column among models with at least five false greens. The column is the evidence field the reply line names, a self-report. The I-9 status column was never cited. Exploratory.

effects here survive a rewording, replicate on fresh firms and across developers, and the largest involves no change to the prompt. Whether a proxy-based `pass` is an error depends on the standard (§3): it is a reasonable triage guess, and a claim made without the record under the standard a monitoring product implies.

9 Limitations

- **No human expert labels or baseline.** No attorney validated any label; the generated labels are re-derived by an AI-written checker sharing the generator's standard, and the anchor labels are qualified by two AI audits. A blind labelling kit is released.
- **A narrow, synthetic generator.** Four rule families, whole-export withholding, five proxy columns chosen by us; no partial records, scanned documents or OCR. Two rules simplify the law (a timely extension can allow 240 more days before reverification; "business day" is not defined in the handbook).
- **Statistics.** Study 3 has 18 clusters, its I-9 effect three firms (Study 3c adds ten), and pooled estimates weight seven particular models equally. Cells were run once.
- **Model coverage is narrow.** No closed OpenAI or Anthropic model was evaluated, the study using no funds beyond cloud credits; the follow-ups use only the three Google models; several endpoints are previews that will be retired.
- **One domain.** We conjecture, and do not show, that the failure generalises.

Ethics, competing interests and AI use

Every record is synthetic. A high score does not show a system fit for use, and the labels are not a training set. The author works for Settle AI Inc., which builds commercial compliance software; the anchor labels and the rulebook derive from its engine, which is not evaluated, and the audits that weaken those labels are reported in full. No model provider was consulted. Anthropic's Claude wrote the code, ran the experiments and drafted the text; separate agents performed the first label audit and wrote the reference checker, and Gemini 3.1 Pro performed the second audit, all reported as AI checks. Every number is recomputed from released artifacts by `scripts/verify_paper.py`; the author is responsible for the content. Code is released under Apache-2.0 and data under CC-BY-4.0.

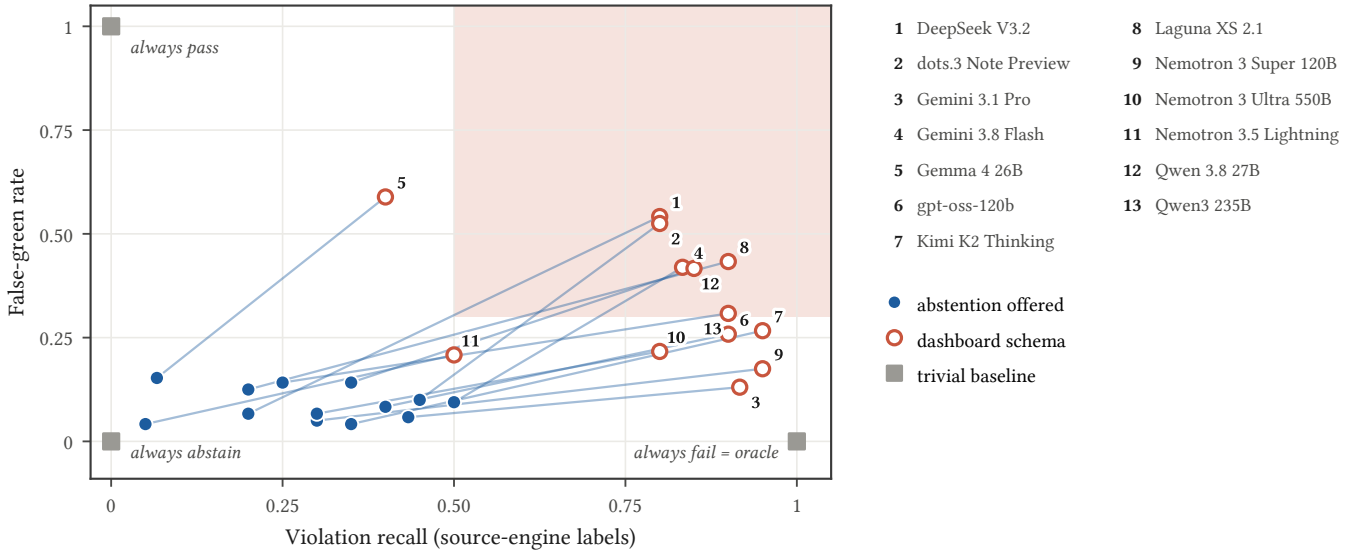


Figure 5. The anchor subset: each model under the offered schema (filled) and the dashboard schema (open), joined by a line; numbers refer to the key. Without the abstention every model moves up and right, buying recall with false greens; the shaded region marks high recall with many false greens. Recall is measured against the source engine's contested violation labels. Grey squares are trivial baselines; always_fail shares the oracle's point because neither axis penalises a fail on an unassessable pair.

References

Afsharrad, A. and Mousavi, S. S. (2026). ImmigrationReason: A Structured Dataset of U.S. Immigration Appeals for Legal Reasoning Research. arXiv preprint arXiv:2608.20391. https://arxiv.org/abs/2608.20391

Aly, R., Guo, Z., Schlichtkrull, M., Thorne, J., Vlachos, A., Christodoulopoulos, C., Cocarascu, O. and Mittal, A. (2021). FEVEROUS: Fact Extraction and VERification Over Unstructured and Structured information. NeurIPS 2021 Datasets and Benchmarks Track. https://arxiv.org/abs/2106.05707

Amaral, O., Abualhaija, S., Torre, D., Sabetzadeh, M. and Briand, L. C. (2022). AI-enabled Automation for Completeness Checking of Privacy Policies. IEEE Transactions on Software Engineering 48(11). https://doi.org/10.1109/TSE.2021.3124332

Chalkidis, I., Jana, A., Hartung, D., Bommarito, M., Androutsopoulos, I., Katz, D. M. and Aletras, N. (2022). LexGLUE: A Benchmark Dataset for Legal Language Understanding in English. Proceedings of ACL 2022. https://aclanthology.org/2022.acl-long.297

Chow, C. (1970). On optimum recognition error and reject tradeoff. IEEE Transactions on Information Theory 16(1):41-46. https://doi.org/10.1109/TIT.1970.1054406

Dahl, M., Magesh, V., Suzgun, M. and Ho, D. E. (2024). Large Legal Fictions: Profiling Legal Hallucinations in Large Language Models. Journal of Legal Analysis 16(1):64-93. https://arxiv.org/abs/2401.01301

Dou, G., Brena, L., Deo, A., Jurayj, W., Zhang, J., Holzenberger, N. and Van Durme, B. (2026). DeonticBench: A Benchmark for Reasoning over Rules. arXiv preprint arXiv:2604.04443. https://arxiv.org/abs/2604.04443

El-Yaniv, R. and Wiener, Y. (2010). On the Foundations of Noise-free Selective Classification. Journal of Machine Learning Research 11:1605-1641. https://jmlr.org/papers/v11/el-yaniv10a.html

Fei, Z., Shen, X., Zhu, D., Zhou, F., Han, Z., Zhang, S., Chen, K., Shen, Z. and Ge, J. (2024). LawBench: Benchmarking Legal Knowledge of Large Language Models. Proceedings of EMNLP 2024. https://aclanthology.org/2024.emnlp-main.452

Fu, Y., Kang, Y. and Zhao, Y. (2026). CalBrief: A Pilot Diagnostic Benchmark for Evidence-Calibrated Scientific Briefing with Large Language Models. arXiv preprint arXiv:2606.27383. https://arxiv.org/abs/2606.27383

Geifman, Y. and El-Yaniv, R. (2017). Selective Classification for Deep Neural Networks. Advances in Neural Information Processing Systems 30. https://arxiv.org/abs/1705.08500

Geirhos, R., Jacobsen, J.-H., Michaelis, C., Zemel, R., Brendel, W., Bethge, M. and Wichmann, F. A. (2020). Shortcut Learning in Deep Neural Networks. Nature Machine Intelligence 2:665-673. https://doi.org/10.1038/s42256-020-00257-z

Guha, N., Nyarko, J., Ho, D. E., Ré, C., et al. (2023). LegalBench: A Collaboratively Built Benchmark for Measuring Legal Reasoning in Large Language Models. NeurIPS 2023 Datasets and Benchmarks Track. https://arxiv.org/abs/2308.11462

Guldimann, P., Spiridonov, A., Staab, R., Jovanović, N., Vero, M., Vechev, V., Gueorguieva, A.-M., Balunović, M., Konstantinov, N., Bielik, P., Tsankov, P. and Vechev, M. (2024). COMPL-AI Framework: A Technical Interpretation and LLM Benchmarking Suite for the EU Artificial Intelligence Act. arXiv preprint arXiv:2410.07959. https://arxiv.org/abs/2410.07959

Guo, K. H., Yan, C., Baidya, A., Brown, K., Gao, X., Xiong, J., Yin, Z. and Malin, B. A. (2026). CLEAR: Revealing How Noise and Ambiguity Degrade Reliability in LLMs for Medicine. arXiv preprint arXiv:2605.01011. https://arxiv.org/abs/2605.01011

Hendrycks, D., Burns, C., Chen, A. and Ball, S. (2021). CUAD: An Expert-Annotated NLP Dataset for Legal Contract Review. NeurIPS 2021 Datasets and Benchmarks Track. https://arxiv.org/abs/2103.06268

Joren, H., Zhang, J., Ferng, C.-S., Juan, D.-C., Taly, A. and Rashtchian, C. (2024). Sufficient Context: A New Lens on Retrieval Augmented Generation Systems. arXiv preprint arXiv:2411.06037. https://arxiv.org/abs/2411.06037

Kadavath, S., Conerly, T., Askell, A., Henighan, T., Drain, D., Perez, E., et al. (2022). Language Models (Mostly) Know What They Know. arXiv preprint arXiv:2207.05221. https://arxiv.org/abs/2207.05221

Kalai, A. T., Nachum, O., Vempala, S. S. and Zhang, E. (2025). Why Language Models Hallucinate. arXiv preprint arXiv:2509.04664. https://arxiv.org/abs/2509.04664

Kirichenko, P., Ibrahim, M., Chaudhuri, K. and Bell, S. J. (2025). AbstentionBench: Reasoning LLMs Fail on Unanswerable Questions. arXiv preprint arXiv:2506.09038. https://arxiv.org/abs/2506.09038

Koreeda, Y. and Manning, C. D. (2021). ContractNLI: A Dataset for Document-level Natural Language Inference for Contracts. Findings of EMNLP 2021. https://aclanthology.org/2021.findings-emnlp.164

Lee, G., Chay, W., Cho, S. and Choi, E. (2024). TrustSQL: Benchmarking Text-to-SQL Reliability with Penalty-Based Scoring. arXiv preprint arXiv:2403.15879. https://arxiv.org/abs/2403.15879

Lee, G., Hwang, H., Bae, S., Kwon, Y., Shin, W., Yang, S., Seo, M., Kim, J.-Y. and Choi, E. (2022). EHRSQL: A Practical Text-to-SQL Benchmark for Electronic Health Records. NeurIPS 2022 Datasets and Benchmarks Track. https://arxiv.org/abs/2301.07695

Luo, Y., Jiang, Y., Xie, Q., Lan, L., Cong, L. W., Rao, A. and Song, Y. (2026). ReguSim: Evaluating LLM Agent Rule Grounding in Financial Compliance. arXiv preprint arXiv:2608.19974. https://arxiv.org/abs/2608.19974

Madhusudhan, N., Madhusudhan, S. T., Yadav, V. and Hashemi, M. (2025). Do LLMs Know When to NOT Answer? Investigating Abstention Abilities of Large Language Models. Proceedings of COLING 2025. https://aclanthology.org/2025.coling-main.627

Magesh, V., Surani, F., Dahl, M., Suzgun, M., Manning, C. D. and Ho, D. E. (2025). Hallucination-Free? Assessing the Reliability of Leading AI Legal Research Tools. Journal of Empirical Legal Studies 22(2). https://doi.org/10.1111/jels.12413

Marino, B., Hunter, R., Schnabl, C., Jamali, Z., Kalpakos, M. E., Kashyap, M., Hinton, I., Hanson, A., Nazir, M., Steffek, F., Wen, H. and Lane, N. D. (2025). AIReg-Bench: Benchmarking Language Models That Assess AI Regulation Compliance. arXiv preprint arXiv:2510.01474. https://arxiv.org/abs/2510.01474

Rajpurkar, P., Jia, R. and Liang, P. (2018). Know What You Don't Know: Unanswerable Questions for SQuAD. Proceedings of ACL 2018. https://aclanthology.org/P18-2124

Ravichander, A., Black, A. W., Wilson, S., Norton, T. and Sadeh, N. (2019). Question Answering for Privacy Policies: Combining Computational and Legal Perspectives. Proceedings of EMNLP-IJCNLP 2019. https://aclanthology.org/D19-1500

Röttger, P., Hofmann, V., Pyatkin, V., Hinck, M., Kirk, H. R., Schütze, H. and Hovy, D. (2024). Political Compass or Spinning Arrow? Towards More Meaningful Evaluations for Values and Opinions in Large Language Models. Proceedings of ACL 2024. https://aclanthology.org/2024.acl-long.816

Sarkar, D. (2026). A Four-Axis Trustworthiness Benchmark for LLM-as-Judge in Principle-Based Regulation. KDD 2026 Workshop on Secure and Trustworthy Large Language Models. arXiv:2608.14329. https://arxiv.org/abs/2608.14329

Sclar, M., Choi, Y., Tsvetkov, Y. and Suhr, A. (2024). Quantifying Language Models' Sensitivity to Spurious Features in Prompt Design or: How I learned to start worrying about prompt formatting. ICLR 2024. https://arxiv.org/abs/2310.11324

Shi, F., Chen, X., Misra, K., Scales, N., Dohan, D., Chi, E., Schärli, N. and Zhou, D. (2023). Large Language Models Can Be Easily Distracted by Irrelevant Context. ICML 2023. https://arxiv.org/abs/2302.00093

Shportun, N. (2026). ImmigrationQA: A Source-Grounded Dataset and Small-Model Adaptation for U.S. Immigration Law. arXiv preprint arXiv:2605.30589. https://arxiv.org/abs/2605.30589

Solozobov, O. (2026). DEMM-Bench: A Cross-Regime Benchmark for Agent-Runtime Governance-Evidence Sufficiency. arXiv preprint arXiv:2606.20634. https://arxiv.org/abs/2606.20634

Tallam, K. (2026). Partial Evidence Bench: Benchmarking Authorization-Limited Evidence in Agentic Systems. arXiv preprint arXiv:2605.05379. https://arxiv.org/abs/2605.05379

Tam, Z. R., Wu, C.-K., Tsai, Y.-L., Lin, C.-Y., Lee, H. and Chen, Y.-N. (2024). Let Me Speak Freely? A Study On The Impact Of Format Restrictions On Large Language Model Performance. Proceedings of EMNLP 2024: Industry Track. https://aclanthology.org/2024.emnlp-industry.91

Theologitis, M., Dammu, P. P. S., Shah, C. and Suciu, D. (2026). ClaimDB: A Fact Verification Benchmark over Large Structured Data. ACL 2026. arXiv:2601.14698. https://arxiv.org/abs/2601.14698

Thorne, J., Vlachos, A., Christodoulopoulos, C. and Mittal, A. (2018). FEVER: a Large-scale Dataset for Fact Extraction and VERification. Proceedings of NAACL-HLT 2018. https://aclanthology.org/N18-1074

Turpin, M., Michael, J., Perez, E. and Bowman, S. R. (2023). Language Models Don't Always Say What They Think: Unfaithful Explanations in Chain-of-Thought Prompting. NeurIPS 2023. https://arxiv.org/abs/2305.04388

Wang, B., Gao, Y., Li, Z. and Lou, J.-G. (2023). Know What I don't Know: Handling Ambiguous and Unknown Questions for Text-to-SQL. Findings of ACL 2023. https://aclanthology.org/2023.findings-acl.352

Wang, J., Wang, M., Zhou, Y., Xing, Z., Liu, Q., Xu, X., Zhang, W. and Zhu, L. (2025). LLM-based HSE Compliance Assessment: Benchmark, Performance, and Advancements. arXiv preprint arXiv:2505.22959. https://arxiv.org/abs/2505.22959

Wang, N. X. R., Mahajan, D., Danilevsky, M. and Rosenthal, S. (2021). SemEval-2021 Task 9: Fact Verification and Evidence Finding for Tabular Data in Scientific Documents (SEM-TAB-FACTS). Proceedings of SemEval-2021.

https://aclanthology.org/2021.semeval-1.39

Wen, B., Yao, J., Feng, S., Xu, C., Tsvetkov, Y., Howe, B. and Wang, L. L. (2025). Know Your Limits: A Survey of Abstention in Large Language Models. Transactions of the Association for Computational Linguistics. https://aclanthology.org/2025.tacl-1.26

Yang, Y., Kim, I.-J. and Yoon, D. (2026). PASTA: A Scalable Framework for Multi-Policy AI Compliance Evaluation. Proceedings of CHI 2026. https://doi.org/10.1145/3772318.3790966

Yin, Z., Sun, Q., Guo, Q., Wu, J., Qiu, X. and Huang, X. (2023). Do Large Language Models Know What They Don't Know? Findings of ACL 2023. https://aclanthology.org/2023.findings-acl.551

Zhang, J. and El-Gohary, N. M. (2016). Semantic NLP-Based Information Extraction from Construction Regulatory Documents for Automated Compliance Checking. Journal of Computing in Civil Engineering 30(2). https://doi.org/10.1061/(ASCE)CP.1943-5487.0000346

Zheng, L., Guha, N., Anderson, B. R., Henderson, P. and Ho, D. E. (2021). When Does Pretraining Help? Assessing Self-Supervised Learning for Law and the CaseHOLD Dataset. ICAIL 2021. https://arxiv.org/abs/2104.08671

Zhou, R., Hua, W., Pan, L., Cheng, S., Wu, X., Yu, E. and Wang, W. Y. (2025). RuleArena: A Benchmark for Rule-Guided Reasoning with LLMs in Real-World Scenarios. Proceedings of ACL 2025. https://aclanthology.org/2025.acl-long.27

—

A Reproduction

```
python3 tests/test_benchmark.py; python3 tests/test_llm_harness.py; python3 tests/test_freetier.py
python3 scripts/sweep.py --firms 30            # Table 5 (trivial baselines)
python3 scripts/score_views.py                  # anchor study, label views, Table 10
python3 scripts/score_studies.py                # repeated-run study and Study 2
python3 scripts/score_study3.py                 # Study 3, the declared analysis
python3 scripts/study3_exploratory.py           # Section 6.3 (exploratory)
python3 scripts/labelcheck_compare.py           # generated labels vs the reference checker
python3 scripts/fill_results.py                 # regenerates every table in this paper
python3 scripts/make_figures.py                 # regenerates every chart
python3 scripts/verify_paper.py                 # recomputes every number in the text
```

No step makes an API call: every score is rebuilt from the retained replies in `docs/**/cache/` and `docs/raw/`. Re-running the models uses `scripts/run_free.py`, `scripts/run_vertex.py` and `scripts/run_study3.py`.

B Prompt and protocol

The prompts are frozen in `baselines/prompts.py` at `PROMPT_VERSION` 1.1.0 and reproduced verbatim below. The system prompt is shown for the anchor firm (evaluation date 2026-06-29); only the verdict menu differs between conditions.

System prompt, `abstain_offered`:

```
You are auditing an employer's immigration-compliance records.

You will be given:
  1. the employer's source data exports, verbatim, as CSV
  2. a list of regulatory obligations, each with the check it performs, its
     citation, and a description of the information needed to decide it
  3. a list of (obligation, entity) pairs to return a verdict for

Evaluate each pair against the data actually present in the exports. Today's
date for the purpose of every deadline calculation is 2026-06-29. Do not use the
real current date.

Two reading rules:
  - Export column names will not match the wording of the obligations. Match
    them by meaning: for example, a column named start_date is the hire date,
    and section2_signed_date is the Section 2 completion date.
  - Reference knowledge is not evidence you need from the exports. Calendars
    and federal holidays, regulatory thresholds, and the evaluation date are
    yours to apply. Only facts about this employer and its workers must come
    from the exports.

Return one line per pair, in this exact format, and nothing else:

  <pair_id>        <verdict>        <evidence_field_or_dash>

Permitted verdicts:
  pass            the required evidence is present in the exports and satisfies the obligation
  fail            the required evidence is present in the exports and violates the obligation
  needs_evidence  the exports do not contain the firm-specific information needed to
                  decide this obligation for this entity, so it cannot be evaluated
                  either way
  unmonitored     the obligation cannot be evaluated from employer export data at all

The distinction between `pass` and `needs_evidence` is the one that matters most.
`pass` asserts you checked and the obligation is met. `needs_evidence` says the
data needed to check was never provided. A document that was not provided is not
a document that is in order.

Return a line for every pair. Do not add commentary, headers, or explanation.
```

Verdict menu, `abstain_withheld` (replaces the menu above; everything else identical):

```
Permitted verdicts:
  pass            the obligation is satisfied
  fail            the obligation is violated

Every pair must receive one of these two verdicts.
```

Verdict menu, `abstain_removed_only` (Table 2):

```
Permitted verdicts:
  pass            the required evidence is present in the exports and satisfies the obligation
```

```
  fail            the required evidence is present in the exports and violates the obligation
  unmonitored     the obligation cannot be evaluated from employer export data at all

A document that was not provided is not a document that is in order.
```

Verdict menu, `abstain_closed` (Table 2):

```
Permitted verdicts:
  pass            the required evidence is present in the exports and satisfies the obligation
  fail            the required evidence is present in the exports and violates the obligation

A document that was not provided is not a document that is in order.

Every pair must receive one of these two verdicts.
```

User prompt. Each request continues with the exports relevant to the batch (verbatim CSV), the obligations with their check, threshold, citation and required information, and the twelve pairs to decide. A complete example is in `docs/raw/`.

C Protocol history and run log

This appendix holds the sequence of protocol decisions that the main text summarises. Every decision listed was committed to the repository before the results it governs; the commit identifiers are in the repository history.

A pilot under prompt v1.0 exposed three defects in our own harness, not in the models: obligations sent without their definitions, a mangled "information needed" field, and a definition of `needs_evidence` that a literal reader could apply to every pair. We fixed them in prompt v1.1.0, archived the pilot, and report no pilot number as a result. The main runs then exposed an output cap that cut off reasoning models mid-reply; incomplete replies were purged and re-requested under a larger cap. Both revisions, with the numbers that motivated them, follow.

C.1 The v1.0 pilot and the revision it forced

The protocol above is version 1.1. Version 1.0 was run first, on the same 140 pairs, and completed four of twelve model-condition runs before we stopped it. We report what it found because it changed the protocol, and a reader should be able to judge whether the change was principled.

The pilot showed violation recall of 0.000 for both models whenever the abstention option was offered. On inspection this was mostly the harness, for three reasons. Thirty of the 140 pairs, including 10 of the 20 planted violations, used rule identifiers absent from the version 1.0 rulebook and reached the prompt with no obligation text at all. The evidence list shown for every obligation had been damaged by our extraction into run-together tokens that matched no export column, and it named reference data no export contains, such as a federal-holiday calendar. And version 1.0 defined `needs_evidence` as a required field being absent, so a model that followed the prompt literally was obliged to abstain. In the clearest case, a Section 2 completed eight business days late, both dates were in the I-9 export, and the model still answered `needs_evidence`.

Version 1.1 fixes these defects and changes nothing else. Every pair's rule is defined. The evidence description is the manifest's own prose. `needs_evidence` is defined by the firm's records rather than by field names. The same two reading rules are added to both ablation conditions. The verdict menus are otherwise unchanged. The evaluated pairs are identical, fixed in a file before either version ran, so any difference between the pilot and the main run is attributable to the prompt. The pilot's raw replies and scores are retained in the repository.

The first attempt at the version 1.1 run exposed one further defect, in the harness rather than the prompt. We had capped each reply at 8,400 output tokens. Reasoning models spend most of a reply thinking before they answer, and 46 of the first 84 batch replies were cut off or came back empty before any verdict was written. The pairs in those batches would have scored as misses, depressing recall for reasons that had nothing to do with the models. We raised the cap to 32,000 tokens, which is below every evaluated model's own limit, and re-requested only the affected batches. The 38 replies that had finished within the lower cap were kept, because a cap that was never reached did not affect them. Truncated or empty replies are no longer cached, and the parser now counts a pair as unparseable only if it is never answered, not each time a model mentions it while reasoning. The affected attempt is archived intact beside the pilot.

One pilot observation did not depend on the defects. When forced to choose between `pass` and `fail`, both pilot models called most unassessable obligations violations rather than compliant, with false-green rates of 0.117 and 0.075. We report version 1.1 as the result and the pilot only as the account of how the protocol reached its final form.

C.2 Model substitutions

Free-tier availability was decided by provider capacity: of 15 free chat models screened, only 6 returned a usable answer to a two-pair request (5 were rate-limited, 2 refused with a permissions error, 1 returned an upstream failure and 1 an empty

completion). The study was declared with six free models. After four of the twelve model-condition runs had completed, OpenRouter withdrew the free versions of both Nex models (N2.5 Pro and N2.5 Mini); the paid versions remain, and paying for them was outside the free-tier studies' zero-cost constraint. Three batches of Nex N2.5 Pro replies had been collected; they cover too few pairs to score and are reported as an incomplete run, not as a result. Dropping both models would have left four, three of them from one provider family (NVIDIA Nemotron). We therefore replaced them with Qwen 3.8 27B and Gemma 4 31B, chosen for provider diversity among the models still free whose output limit exceeds the 32,000-token cap, with Inkling named in advance as the substitute should either fail the same two-pair screen the original models passed. Inkling then turned out to refuse API calls outright (it is served only inside agent applications), so before any replacement reply existed the fallback was changed to dots.3 Note Preview, the one remaining free model from a provider family not already in the set. A transient upstream rate limit is not treated as a screen failure. Both replacements then returned upstream rate limits on every screen attempt for over an hour, so a cut-off was added, again before any replacement had answered a benchmark batch: when the sweep reaches a replacement it screens it once more, and a model that has still never passed yields its slot to the fallback. The fallback fills at most one slot; a second unscreenable replacement is dropped and reported as such. Every screen attempt is logged (`docs/replacement_screen_2026-09-30.json`). The replacements and the fallback rule were committed on 30 September 2026, before any reply from either model existed; as with the label views, the commit history is the evidence of that ordering. The replacements run under the identical prompt, subset, batch size, and output cap.

C.3 Throughput on free tiers

The OpenRouter free tier allowed 50 requests a day across all free models and 20 a minute. At batch 12, the Study 1 subset costs 12 calls per model and condition, and each Study 2 run (six generated firms) costs 24. The runner paces itself under the per-minute limit, waits patiently on upstream rate limits (which were not charged against the daily allowance), stops cleanly when the daily ceiling is reached, and resumes from a checkpoint. Every model response is retained so that any reported number can be traced back to output.

C.4 Departures from the delivery-failure rule

First, Nemotron 3.5 Lightning's second run never finished normally: its host repeatedly cut replies off at a ten-minute limit. Under a cut-off rule committed before the final quota window, once every other run had finished, the run was given one pass that tried each remaining batch up to three times and was then recorded with what it had answered: 68 of 140 pairs. Second, during that pass one reply ended with an upstream error *after* delivering all twelve verdict lines. The harness accepted it because the reply was not empty, contrary to the rule above. We keep it, since its answer is complete, and the harness has since been corrected to re-request such replies. It is the only reply in the study that ended without a normal stop.

C.5 Study 3 operations

Study 3 ran as one process per model and arm, then, for the four slowest models, one per condition and finally, for gpt-oss-120b, four firm shards per condition, to finish within a day. Each process kept its own cache file. A shard began from a copy of its condition's cache, so batches already answered were reused, not requested again, and no request was sent twice with different results kept. No run ended incomplete. Qwen3 235B's requests were capped at its served limit of 16,384 tokens, as declared. Before the declaration, the four open models were sent only a one-word availability probe. The free-tier Study 2 sweep was stopped once Study 3 was declared; its one completed model's replies are retained in `docs/study2/` and not reported.

C.6 Source scoping and the anchor subset

Source scoping. Free tiers cap tokens per minute, so each call carries only the exports whose regime the batch's obligations can be evaluated against, with the HR roster always included; every run uses batches of 12 pairs. Scoping changes the prompt, so it is part of the protocol. The prompt names the exports not sent and tells the model not to infer that an absent field exists in one of them, and an obligation of an unclassified regime falls back to all sources. Scoping is identical across conditions, so it cannot bias a contrast between them; we did not run without it. One scoping defect was found and fixed before Study 3 (Appendix E).

Model set. Six free models were declared; two were withdrawn from the free tier mid-study and replaced under rules committed before any replacement answered a batch (C.2).

The declared evaluation subset. The free-tier budget (50 calls a day) does not cover the full anchor firm, so Study 1 evaluates a **declared stratified subset** of 140 pairs, fixed by a seed and frozen before any model was scored: **every planted violation** and 120 of the 201 withheld pairs, sampled in proportion to regime. We never present a subset rate as a full-firm rate.

Table 5. Trivial baselines. Across 31 firms (the anchor and 30 generated), averaged over the firms in which each rate is defined.

System	False green	Violation recall	Precision	False alarm	Abstention F1
`always_pass`	1.000	0.000	—	0.000	0.000
`always_abstain`	0.000	0.000	—	1.000	0.717
`always_fail`	0.000	1.000	0.170	1.000	0.000
`oracle`	0.000	1.000	1.000	0.000	1.000

A dash marks a rate undefined for that system: `always_pass` and `always_abstain` never predict a violation, so their precision has no denominator.

Table 6. All thirteen models on the anchor subset.

Model	Developer	Runs	Answered (offered)	FGR offered	FGR offered, answered only	Recall offered (of 20)	FGR dashboard	Change significant?
dots.3 Note Preview	dots-studio	1	140/140	0.100	0.100	0.450	0.525	yes (p = 0.003)
Nemotron 3 Super 120B	NVIDIA	1	140/140	0.050	0.050	0.300	0.175	yes (p = 0.012)
Nemotron 3 Ultra 550B	NVIDIA	1	140/140	0.067	0.067	0.300	0.217	yes (p = 0.003)
Nemotron 3.5 Lightning	NVIDIA	1	92/140	0.042	0.059	0.050	0.208	yes (p = 0.012)
Laguna XS 2.1	Poolside	1	124/140	0.125	0.135	0.200	0.433	yes (p = 0.003)
Qwen 3.8 27B	Alibaba	1	140/140	0.142	0.142	0.350	0.417	yes (p = 0.003)
Gemini 3.1 Pro	Google	3 (mean)	140/140 (min)	0.058	0.058	0.433	0.131	no (p = 1.000)
Gemini 3.8 Flash	Google	3 (mean)	140/140 (min)	0.094	0.094	0.500	0.419	yes (p = 0.012)
Gemma 4 26B	Google	3 (mean)	140/140 (min)	0.153	0.153	0.067	0.589	yes (p = 0.012)
gpt-oss-120b	OpenAI	1	140/140	0.083	0.083	0.400	0.258	not tested
DeepSeek V3.2	DeepSeek	1	140/140	0.067	0.067	0.200	0.542	not tested
Qwen3 235B	Alibaba	1	128/140	0.142	0.155	0.250	0.308	not tested
Kimi K2 Thinking	Moonshot	1	140/140	0.042	0.042	0.350	0.267	not tested

FGR over the 120 unassessable pairs of the 140-pair anchor subset; unanswered pairs cannot be false greens, so the answered-only column guards against coverage effects. Change: offered to dashboard, paired on pairs answered in both; Holm across the six single-run models, and across 24 contrasts for the three repeated-run models. The last four models were added with Study 3 and run once, descriptively. Recall is against the source engine's violation labels (Section 4.4).

Component	Obligations exercised	Regimes	Pairs	Label source
Rulebook	194	10	—	Cited regulation per obligation
Anchor firm	112	8	221	Source engine, audited twice
Anchor subset (Section 7)	76	8	140	As above
Generator	4	2	42 per firm	By construction, independently re-derived

C.7 Generator v1, withdrawn

A first version emitted a label for every pair in the 194-obligation rulebook but rendered deciding data for only four rule families, and applied rules to entities they cannot govern. A blind check caught this before any result on generated firms was reported: an AI agent given only the records and rule text for 60 stratified pairs agreed with its labels on 8. Every figure from that version, including a released-set size of 58,961 pairs, is withdrawn (`docs/study2_v1_invalid/`). The same check on the current generator agreed on all 60.

C.8 Studies 3b and 3c

Both were declared before any call (`docs/STUDY3B_PARAPHRASE.md`, `docs/STUDY3C_DECLARATION.md`) and run on the three Google models through Vertex AI, one process per model, rendering and condition; Gemini 3.1 Pro and 3.8 Flash were later split into three firm shards each, every shard starting from its condition's cache so no answered batch was requested twice.

D The anchor firm's labels and per-model results

D.1 Trivial baselines and all thirteen models

`always_pass` raises no false alarm while reporting all 723 unassessable pairs as satisfied; `always_abstain` scores 0.000 on the false-green rate and 0.717 abstention F1 (0.953 on the anchor firm) while detecting nothing; `always_fail` and the oracle both score 0 on the false-green rate and 1 on recall, which is why the false-alarm rate must be reported too.

D.2 The rulebook, the anchor firm and the released data

The rulebook's 194 obligations fall in ten regimes: I-9 and E-Verify (28), Labor Condition Application (23), Public Access File (15), wage and prevailing wage (21), H-1B status (16), I-129 supplement (13), L-1 (21), F-1 OPT and STEM OPT (15), site-visit readiness (14), and deadlines and penalties (28). The source manifest lists 196 entries; two rules appear in two regimes and are counted once. The anchor firm also refers to 17 rules by the engine's identifiers, mapped to manifest rules through its published alias table. The anchor firm's eleven exports hold 93 data rows for twelve workers, nine on H-1B status.

The released benchmark comprises 31 firms (the anchor and 30 generated at seeds 1-30), 1,481 pairs, 723 of them unassessable (48.8%), and 238 planted violations; further firms are available at any seed.

D.3 Known problems with the anchor labels

The anchor firm's labels were written as the acceptance gate for a compliance engine, not generated, so they are not "fixed by construction" in the sense the generated firms are. Their `fail` means "requires action" in that engine. A review of the 20 violation labels while preparing this paper found the following, all disclosed rather than silently corrected, because the labels predate every result:

- *Duplicates.* Four facts each carry two violation labels under different rules: Mei Tanaka's wage shortfall, Sofia Reyes's worksite, Daniel Cho's E-Verify case, and Hiroshi Yamamoto's petition expiry. The 20 labels describe 16 distinct facts.
- *Not violations on the records.* Five "actual-wage memo missing" labels have no evidence in any export, and five are deadlines that had not passed on the evaluation date. The strict view (D.3) handles both.
- *Contestable plants.* Some plants ignore exceptions a practitioner would check. A roster location outside the LCA's area may be a permitted short-term placement (20 C.F.R. 655.735). A management title against an individual-contributor occupation code is not by itself a material change, since the test is a change in duties (8 C.F.R. 214.2(h)(2)(i)(E)). Helena Voss's reverification label has no reverification data in the export. One certified LCA states an offered wage below its prevailing wage, which the Department of Labor would not certify.

We keep the engine's labels as the headline view because they existed before this study. The strict view, the duplicate count and the generated firms are the checks on them.

The firm's expectation set contains no `pass` labels. Some workers' records conform, but the engine's expectations name only violations and evidence requests, so the anchor firm cannot measure false alarms; the generated firms can (Section 4.2).

Its expectation set contains 221 pairs: 20 planted violations and 201 evidence requests. The 201 arise because most obligations in this domain require evidence an employer export does not contain. A certified LCA does not evidence that the required notice was posted; a roster does not evidence which document a worker presented. The honest output for such a pair is an evidence request. The ten-to-one ratio reflects which obligations the source engine can assess from exports, not a measured property of real employers; we treat the large abstention share as a design choice that mirrors that engine's coverage.

D.4 Label views and the two audits

Two label views, fixed before per-pair results were read. The anchor firm's labels were written as the acceptance gate for a compliance engine, where `fail` means "requires action"; here `fail` means "the records show a violation". Ten of the 20 planted violations sit between those meanings, so we score under two views, fixed in `nofalsegreen/labelviews.py` on 2026-09-24 with each affected pair named. The commit came six minutes after a first, later-discarded attempt at the main run had printed preliminary aggregate rates to its log; no per-pair result had been read, but we cannot exclude that those aggregates influenced the decision.

- **Engine view, the headline.** The labels as authored, which existed before this benchmark did, so nothing decided here can have tuned them.
- **Strict view.** Five "actual-wage memo missing" violations become `needs_evidence`, since no export records the memo as absent. Five deadlines not yet passed on the evaluation date (an LCA and a petition near expiry, an open extension window, an RFE and a NOID not yet due) are removed from scoring and reported as deadline alerts. The remaining ten are breaches that had already occurred.

The strict view's ten violations include three duplicated facts (D.2), so they describe seven distinct facts. Its recall has a denominator of ten, so one pair moves it by 0.1, and it is always reported beside the engine view.

A third view, from a blind audit of the withheld labels. While reading early results we found a withheld label the exports contradict: `i9.section2.employer_signature` is labelled `needs_evidence` because the source engine never maps the column, yet `section2_signed_date` and `section2_examiner_name` are populated for every employee. Because this was noticed in model output, the correction was constrained so that output could not steer it. The protocol (`docs/LABEL_AUDIT_PROTOCOL.md`) was committed before the audit ran; it covers all 120 withheld pairs; the auditor, a separate agent, saw only each pair's obligation text and exports; and a label is overturned only if every required item is present for that entity. The audit overturned 2 of 120, the Section 1 and Section 2 I-9 signatures, both on signed-date columns (the Section 1 call is the weaker: a date is not obviously a signature). In the **audited view** those pairs are removed from scoring, not relabelled. Removing pairs can only remove false greens, so every table in this view states its denominator, and the engine view remains the headline.

A second auditor, and an audit of the violations. Under a protocol addendum committed before it ran, a second auditor from a different developer (Gemini 3.1 Pro, also a model under test, which never saw its own replies) repeated the

Table 7. Single-run models on the anchor subset, offered schema. One row per model on the 140-pair subset, with the trivial baselines as reference rows. The false-alarm column is 0.000 by construction, since the subset has no conforming pair.

Model	False green	FG, answered only	Recall	Precision	False alarm	Abst. P	Abst. R	Abst. F1	Unparse.	Answered
dots.3 Note Preview	**0.100** [0.03, 0.21]	0.100	0.450 [0.20, 0.79]	0.900	0.000	0.906	0.883	0.894	0.000	140/140
Laguna XS 2.1	**0.125** [0.05, 0.24]	0.135	0.200 [0.04, 0.45]	0.667	0.000	0.940	0.783	0.855	0.000	124/140
Nemotron 3 Super 120B	**0.050** [0.01, 0.11]	0.050	0.300 [0.10, 0.61]	0.857	0.000	0.896	0.858	0.877	0.000	140/140
Nemotron 3 Ultra 550B	**0.067** [0.02, 0.14]	0.067	0.300 [0.10, 0.60]	1.000	0.000	0.903	0.933	0.918	0.000	140/140
Nemotron 3.5 Lightning	**0.042** [0.01, 0.09]	0.059	0.050 [0.00, 0.18]	0.500	0.000	0.951	0.642	0.766	0.000	92/140
Qwen 3.8 27B	**0.142** [0.06, 0.25]	0.142	0.350 [0.14, 0.67]	0.875	0.000	0.894	0.842	0.867	0.000	140/140
`always_pass` (ref)	1.000	1.000	0.000	—	0.000	0.000	0.000	0.000	0.000	140/140
`oracle` (ref)	0.000	0.000	1.000	1.000	0.000	1.000	1.000	1.000	0.000	140/140

Table 8. Strict label view. The same runs scored under the strict view (D.3). Alerts: the verdicts given on the five future deadlines the strict view removes.

Model	Condition	FGR, engine	FGR, strict	Recall, engine (of 20)	Recall, strict (of 10)	Alerts: fail / pass / other
dots.3 Note Preview	offered	0.100	0.096	0.45	0.70	2 / 0 / 3
dots.3 Note Preview	dashboard	0.525	0.504	0.80	0.80	3 / 2 / 0
Nemotron 3 Super 120B	offered	0.050	0.048	0.30	0.40	2 / 0 / 3
Nemotron 3 Super 120B	dashboard	0.175	0.168	0.95	0.90	5 / 0 / 0
Nemotron 3 Ultra 550B	offered	0.067	0.064	0.30	0.50	1 / 1 / 3
Nemotron 3 Ultra 550B	dashboard	0.217	0.208	0.80	0.90	2 / 3 / 0
Nemotron 3.5 Lightning	offered	0.042	0.040	0.05	0.10	0 / 1 / 4
Nemotron 3.5 Lightning	dashboard	0.208	0.200	0.50	0.40	3 / 0 / 2
Laguna XS 2.1	offered	0.125	0.120	0.20	0.30	1 / 0 / 4
Laguna XS 2.1	dashboard	0.433	0.416	0.90	0.80	5 / 0 / 0
Qwen 3.8 27B	offered	0.142	0.136	0.35	0.50	2 / 0 / 3
Qwen 3.8 27B	dashboard	0.417	0.400	0.85	0.90	3 / 2 / 0

withheld audit and, blind to the label, judged each violation pair as `violation_shown`, `compliance_shown` or `cannot_tell` (`docs/label_audit_second.json`). On the withheld pairs the two auditors keep the label on 113 of 120. The first overturns 2 and the second 5, with no pair in common; the second's five are three LCA wage pairs, the FDNS itinerary item and the I-9 physical-examination item. On the violations, the second auditor supports neither the five memo absences nor the five future deadlines on the labelled ground, reaching the strict view's correction independently. Of the strict view's ten violations it supports five, covering four distinct facts: a late Section 2, a missed reverification, pay below the required wage, and a lapsed E-Verify action window. It judges four undecidable from the exports (a title change with no occupation code, a worksite move and an LCA area breach with no effective date, an E-Verify referral with no referral date) and one compliant (a U.S. passport holder with no reverification recorded). The false-green rate does not depend on any violation label. Violation recall on the anchor firm does, and should be read as recall against an engine's view of what is actionable, of which roughly half is a breach evident from the exports.

D.5 Per-model results of the single-run models

Six free models were evaluated on the declared 140-pair subset of the anchor firm under prompt v1.1.0, each under both output schemas: Nemotron 3 Ultra 550B, Nemotron 3 Super 120B and Nemotron 3.5 Lightning (NVIDIA), Laguna XS 2.1 (Poolside), Qwen 3.8 27B (Alibaba), and dots.3 Note Preview (dots-studio), the last two replacing withdrawn models (Appendix C.2). Tables are generated from the retained replies (`scripts/fill_results.py`); the coverage and answered-only columns were added after the first results, when the coverage effect of Section 3.3 was found.

Uncertainty. Bracketed figures are 95% percentile intervals from a bootstrap that resamples *obligations* (2,000 draws, fixed seed), since pairs under one obligation share a rule text and records. At 140 pairs most differences *between* models are not resolved, so we do not rank models and list them alphabetically. The comparison the study is built for is *within* a model across schemas: the paired difference on withheld pairs both runs answered, Holm-corrected across the six models. All six remain significant at 0.05 (Table 10).

Offered an abstention, models abstain readily and detect little. False-green rates run from 0.042 to 0.142. The lowest belongs to Lightning, which answered only 92 of 140 pairs; among the withheld pairs it actually answered its rate is 0.059. Violation recall runs from 0.050 to 0.450: models answer `needs_evidence` on violations whose evidence is in the exports. Abstention F1 is between 0.766 and 0.918, so the abstentions mostly land on the right pairs; there are too many of them.

The strict view moves false-green rates by at most 0.021 and raises recall for every model under `abstain_offered`, because

Table 9. Audited label view. The two withheld pairs the first audit overturned are removed; the scored denominator is in every row.

Model	Condition	Unassessable pairs scored	FGR, engine	FGR, audited	FG, audited, answered only
dots.3 Note Preview	offered	118 of 120	0.100	0.085	0.085
dots.3 Note Preview	dashboard	118 of 120	0.525	0.517	0.517
Nemotron 3 Super 120B	offered	118 of 120	0.050	0.034	0.034
Nemotron 3 Super 120B	dashboard	118 of 120	0.175	0.161	0.161
Nemotron 3 Ultra 550B	offered	118 of 120	0.067	0.051	0.051
Nemotron 3 Ultra 550B	dashboard	118 of 120	0.217	0.203	0.203
Nemotron 3.5 Lightning	offered	118 of 120	0.042	0.025	0.036
Nemotron 3.5 Lightning	dashboard	118 of 120	0.208	0.212	0.439
Laguna XS 2.1	offered	118 of 120	0.125	0.110	0.119
Laguna XS 2.1	dashboard	118 of 120	0.433	0.424	0.472
Qwen 3.8 27B	offered	118 of 120	0.142	0.127	0.127
Qwen 3.8 27B	dashboard	118 of 120	0.417	0.407	0.407

Table 10. Offered versus dashboard schema, single-run models.

Model	Unassessable pairs answered in both	FGR offered	FGR dashboard	Change [95% CI]	Holm-adjusted p	Off-schema replies (dashboard)
dots.3 Note Preview	120 of 120	0.100	0.525	+0.425 [+0.26, +0.59]	0.003	0 of 140 answered
Nemotron 3 Super 120B	120 of 120	0.050	0.175	+0.125 [+0.03, +0.24]	0.012	0 of 140 answered
Nemotron 3 Ultra 550B	120 of 120	0.067	0.217	+0.150 [+0.07, +0.27]	0.003	0 of 140 answered
Nemotron 3.5 Lightning	41 of 120	0.042	0.208	+0.463 [+0.12, +0.73]	0.012	0 of 68 answered
Laguna XS 2.1	99 of 120	0.125	0.433	+0.333 [+0.16, +0.51]	0.003	0 of 128 answered
Qwen 3.8 27B	120 of 120	0.142	0.417	+0.275 [+0.13, +0.43]	0.003	0 of 140 answered

FGR columns are over all 120 unassessable pairs. The change, its interval (obligation-cluster bootstrap, 2,000 draws) and the two-sided p-value (Holm across the six models) are computed on the unassessable pairs both runs answered; the bootstrap p-value has a floor of 1/2,000.

Table 11. False-green rate by regime, offered schema. In parentheses, the regime's scored pairs of any label.

Regime	Nemotron 3 Ultra 550B	Nemotron 3 Super 120B	Nemotron 3.5 Lightning	Laguna XS 2.1	Qwen 3.8 27B	dots.3 Note Preview
`deadlines-penalties`	0.000 (6)	0.000 (6)	0.000 (6)	0.000 (6)	0.250 (6)	0.000 (6)
`fdns-sitevisit`	0.000 (7)	0.000 (7)	0.000 (7)	0.000 (7)	0.000 (7)	0.000 (7)
`h1b-status`	0.000 (17)	0.000 (17)	0.000 (17)	0.000 (17)	0.000 (17)	0.000 (17)
`i129-supplement`	0.000 (8)	0.000 (8)	0.000 (8)	0.000 (8)	0.125 (8)	0.000 (8)
`i9-everify`	0.217 (28)	0.174 (28)	0.174 (28)	0.217 (28)	0.304 (28)	0.174 (28)
`lca-dol`	0.100 (33)	0.067 (33)	0.000 (33)	0.100 (33)	0.100 (33)	0.100 (33)
`paf`	0.000 (17)	0.000 (17)	0.059 (17)	0.000 (17)	0.000 (17)	0.000 (17)
`wage-prevailing`	0.000 (24)	0.000 (24)	0.000 (24)	0.389 (24)	0.278 (24)	0.278 (24)

Cells are the false-green rate over the regime's unassessable pairs; in parentheses, the regime's scored pairs of any label.

the five memo pairs most models answered `needs_evidence` stop counting as misses. A `pass` on a deadline days away is the one alert outcome a reviewer would never see: offered an abstention, models gave it on 2 of 30 alerts; without it, on 7 of 30, three of them Nemotron Ultra's.

Removing the two I-9 signature pairs lowers the false-green rate by 0.008 to 0.016 in 11 of the 12 runs; in the twelfth, Lightning without the abstention, it rises by 0.004. It changes neither the ordering of models nor the direction of any condition effect.

Removing the abstention from a dashboard schema raises false greens in every model. The direction is close to built in: a schema with only `pass` and `fail` forces every unassessable pair into one of them. What the design does not fix is the size: it rises by between 0.125 (Nemotron Super) and 0.425 (dots.3), and the lean: five of the six models sent most unassessable pairs to `fail`, while dots.3 sent more to `pass` (63) than to `fail` (57). No model returned an off-schema verdict among the pairs it answered. This condition also redefines `pass` and drops a warning paragraph, so its effect is that of moving to a dashboard schema; Sections 6 and 7 take it apart.

Three regimes, I-9 and E-Verify, prevailing wage, and the LCA rules, hold almost every false green; H-1B status and FDNS site visits have none. Figure 6 shows the same data as a heatmap.

D.6 Further declared analyses

Which failure dominates. Offered an abstention, each model misses between 11 and 19 of the 20 violation labels (counting unanswered pairs as misses) and reports 5 to 17 of the 120 withheld pairs as compliant, so as a rate the missed violation is the larger failure for every model. No reply line named a pair without a verdict; formatting failures were whole replies with

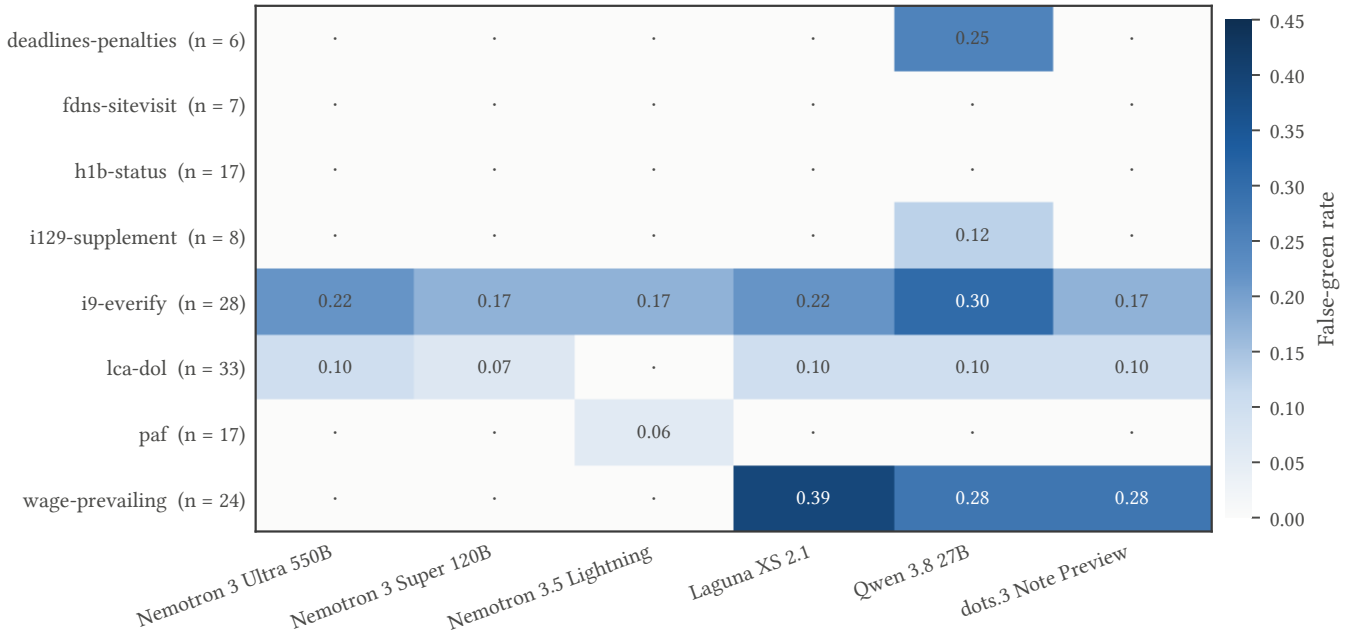


Figure 6. False-green rate by regime and model on the anchor subset, offered schema (the data of Table 11). False greens concentrate in the regimes whose exports carry a populated column next to the missing evidence.

no usable lines, which appear as unanswered pairs (48 and 72 in Lightning's two runs, 16 and 12 in Laguna's). Without the abstention each model misses 1 to 10 violations, and most unassessable pairs become unsupported violations (32 to 99 of 120) or false greens (21 to 63): the withheld condition buys recall by calling unassessable pairs violations.

Where abstention is hardest. The declared hypothesis was that obligations whose evidence is a *document* (a posting notice, a public access file) show more false greens than those whose evidence is a *field* (an I-94 expiry), because a missing document is visible only as a record never expected to exist. Classes were assigned mechanically from each obligation's "information needed" text, committed before any rate was computed by class: 49 document and 71 field pairs among the 120 withheld (`docs/evidence_class_assignment.txt`). *Result:* in 11 of the 12 runs the document rate is higher. Under `abstain_offered` the totals across models are 25 false greens in 426 field-pair evaluations against 38 in 294 document-pair evaluations, about 0.06 against 0.13; the runs share pairs, so we report counts and direction, not a p-value. The one exception, Lightning without the abstention, holds even among pairs it answered (0.575 on field pairs against 0.118 on document pairs).

Qualitative failure cases. To keep excerpts from being the worst lines picked out of thousands, the rule was fixed in code before any was read: for each model, under `abstain_offered`, the first `pass` on a withheld pair whose reply cites an evidence field. Applied as declared, it selected the same line for four of the six free models: a `pass` on the Section 2 employer signature citing `section2_signed_date`, the label the audit later overturned. Four models independently reading a populated signed-date column as a signature suggests the label was at fault (D.3), and the prompt's own example of matching columns by meaning names that column. Applying the rule again while skipping overturned pairs, a step added after the first result was seen, gives the excerpts in Section 7.

E Study 2, six generated firms without proxy columns

Study 2 runs the same protocol on all 252 pairs of six generated firms (Section 4.2), which carry what the anchor firm lacks: conforming controls, a third of them near-miss cases on the legal boundary. Firms were fixed in `nofalsegreen/study2.py` before any call. Results come from the three repeated-run models, three runs each, in the Study 2 rows of Table 13. The declared free-tier sweep was stopped after one model, superseded by Study 3 (Section 6), which runs four models from other developers on fresh firms.

A scoping defect, and what it touches. While preparing Study 3 we found that the scoped prompt for the wage rule never carried a generated firm's `actual_wage_system` export, which the rule needs. In Study 2 every wage pair was therefore sent without it. The 27 withheld wage pairs are unaffected, since their evidence was missing either way. The 27 decidable wage pairs (three firms, labelled `pass` or `fail`) were not decidable from what the models saw, and are excluded from every Study 2 metric. False-green rates and the Table 14 contrasts do not change. Detection and false-alarm figures do, and one reading in an earlier version of this paper was wrong: we attributed Gemini 3.1 Pro's near-miss false alarms (0.162) to caution

hris_roster_export.csv (first two workers; seed 1018, which withholds `i9_record_export`)

Plain

employeeNumber	visaType	jobTitle	hireDate	employmentStatus	…
SYN-1001	H-1B	Computer Systems Analysts	2024-11-05	Active	
SYN-1002	H-1B	Computer Systems Analysts	2022-02-03	Active	

Proxy

employeeNumber	visaType	hireDate	…	i9Status	onboardingCompletedDate	workAuthValidThrough	annualBaseSalary	payBandMidpoint
SYN-1001	H-1B	2024-11-05		Complete	2024-11-02	2027-04-29	118800	106800
SYN-1002	H-1B	2022-02-03		Complete	2022-01-30	2028-11-05	118800	108000

Figure 7. Study 3 firm 1018 as generated, rendered plain and with proxy columns. Both withhold the I-9 export, so every I-9 pair is unassessable in both; the proxy rendering adds five roster columns (shaded); every label and every other export is identical.

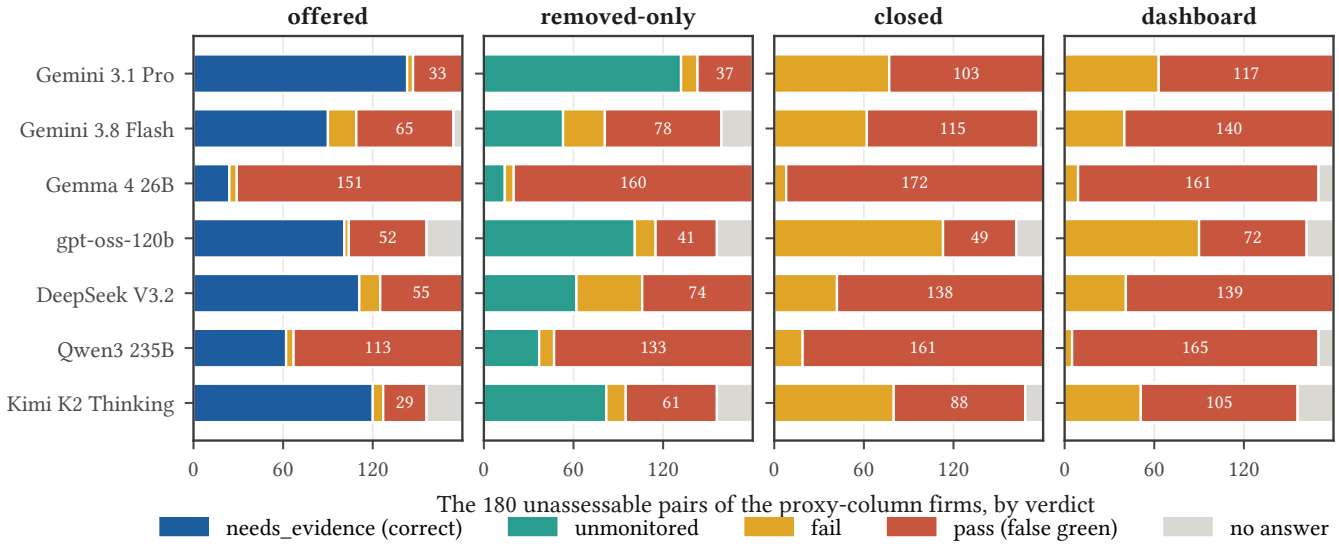


Figure 8. Study 3, proxy-column firms: where the 180 unassessable pairs go under each schema. Removing needs_evidence alone sends many pairs to unmonitored; removing every non-committal verdict sends them to fail or pass, in proportions that differ by model.

Table 12. Schema contrasts on plain and proxy-column firms (exploratory).

Change of schema	Plain firms (exploratory)	Proxy firms (declared test)
Abstention: offered → closed	+0.093 [+0.04, +0.16], p < 0.001 unadjusted	+0.259 [+0.20, +0.33], Holm p = 0.002
Definition and warning: closed → dashboard	+0.035 [+0.01, +0.06], p = 0.008 unadjusted	+0.084 [+0.04, +0.12], Holm p = 0.002

Pooled over the seven models with equal weight; same pairs, clusters and bootstrap draws as the declared analysis. The plain-firm column was computed after the results and is not part of any declared family.

about evidence that was present. They were requests for the missing actual-wage export, which was correct. The defect is fixed for Study 3.

Schema contrasts. On these firms no contrast survives correction (Table 14), consistent with firms that carry no proxy columns.

Generated firms are easier. With abstention offered, the three models report 0.021 to 0.071 of unassessable pairs as compliant, below their anchor-firm rates. Gemini 3.1 Pro and 3.8 Flash detect every violation and raise no false alarm on a conforming control, near-miss controls included, in three of the four schemas. Gemma 4 detects 0.346 of violations and, with abstention offered, raises 0.074 near-miss false alarms. The generated exports are clean, with no populated column next to a missing one, which is the gap Study 3 fills. Excluding the reverification pairs, whose violation labels assume no extension was filed (Section 9), changes no conclusion; no false green fell on them.

F Study 3 details and repeated-run tables

Repeats are close. Across the 24 model, study and schema cells, the three runs' false-green rates differ by at most 0.142, and by 0.06 or less in 21 of them. That bounds run-to-run variation on one host at temperature 0; it says nothing about variation across hosts, so it does not license the single runs of Section 7.

Table 13. The three repeated-run models, both studies, four schemas. Mean of three runs, with the minimum and maximum in brackets.

Model	Study	Schema	False green	FG, answered only	Recall	False alarm, near-miss	Unassessable pairs answered pass / fail / needs_ev. / unmon.	Min answered
Gemini 3.1 Pro	1	offered	0.058 [0.058, 0.058]	0.058	0.433 [0.400, 0.450]	—	7 / 1 / 101 / 11	140/140
Gemini 3.1 Pro	1	removed-only	0.050 [0.025, 0.067]	0.050	0.550 [0.500, 0.600]	—	6 / 5 / 0 / 109	140/140
Gemini 3.1 Pro	1	closed	0.050 [0.033, 0.067]	0.050	0.950 [0.950, 0.950]	—	6 / 114 / 0 / 0	140/140
Gemini 3.1 Pro	1	dashboard	0.131 [0.125, 0.142]	0.131	0.917 [0.850, 0.950]	—	16 / 104 / 0 / 0	140/140
Gemini 3.1 Pro	2	offered	0.021 [0.016, 0.032]	0.021	1.000 [1.000, 1.000]	0.000 [0.000, 0.000]	3 / 0 / 123 / 0	225/252
Gemini 3.1 Pro	2	removed-only	0.056 [0.024, 0.071]	0.056	1.000 [1.000, 1.000]	0.025 [0.000, 0.037]	7 / 26 / 0 / 93	225/252
Gemini 3.1 Pro	2	closed	0.061 [0.056, 0.064]	0.061	1.000 [1.000, 1.000]	0.000 [0.000, 0.000]	8 / 118 / 0 / 0	225/252
Gemini 3.1 Pro	2	dashboard	0.061 [0.056, 0.064]	0.061	1.000 [1.000, 1.000]	0.000 [0.000, 0.000]	8 / 118 / 0 / 0	225/252
Gemini 3.8 Flash	1	offered	0.094 [0.092, 0.100]	0.094	0.500 [0.500, 0.500]	—	11 / 0 / 96 / 12	140/140
Gemini 3.8 Flash	1	removed-only	0.156 [0.125, 0.183]	0.160	0.783 [0.750, 0.850]	—	19 / 18 / 0 / 79	128/140
Gemini 3.8 Flash	1	closed	0.242 [0.208, 0.283]	0.245	0.767 [0.600, 0.850]	—	29 / 89 / 0 / 0	127/140
Gemini 3.8 Flash	1	dashboard	0.419 [0.375, 0.458]	0.433	0.833 [0.800, 0.850]	—	50 / 66 / 0 / 0	127/140
Gemini 3.8 Flash	2	offered	0.026 [0.016, 0.032]	0.027	1.000 [1.000, 1.000]	0.000 [0.000, 0.000]	3 / 4 / 117 / 0	219/252
Gemini 3.8 Flash	2	removed-only	0.037 [0.032, 0.048]	0.037	1.000 [1.000, 1.000]	0.000 [0.000, 0.000]	5 / 62 / 0 / 59	225/252
Gemini 3.8 Flash	2	closed	0.156 [0.127, 0.182]	0.156	1.000 [1.000, 1.000]	0.000 [0.000, 0.000]	20 / 106 / 0 / 0	225/252
Gemini 3.8 Flash	2	dashboard	0.167 [0.167, 0.167]	0.167	1.000 [1.000, 1.000]	0.000 [0.000, 0.000]	21 / 105 / 0 / 0	225/252
Gemma 4 26B	1	offered	0.153 [0.150, 0.158]	0.153	0.067 [0.050, 0.100]	—	18 / 0 / 76 / 26	140/140
Gemma 4 26B	1	removed-only	0.183 [0.175, 0.200]	0.220	0.067 [0.050, 0.100]	—	22 / 0 / 0 / 79	116/140
Gemma 4 26B	1	closed	0.600 [0.525, 0.667]	0.682	0.400 [0.400, 0.400]	—	72 / 33 / 0 / 0	122/140
Gemma 4 26B	1	dashboard	0.589 [0.567, 0.617]	0.635	0.400 [0.400, 0.400]	—	71 / 41 / 0 / 0	128/140
Gemma 4 26B	2	offered	0.071 [0.071, 0.071]	0.071	0.346 [0.296, 0.370]	0.074 [0.074, 0.074]	9 / 0 / 116 / 1	225/252
Gemma 4 26B	2	removed-only	0.071 [0.071, 0.071]	0.071	0.370 [0.333, 0.407]	0.111 [0.111, 0.111]	9 / 0 / 0 / 117	225/252
Gemma 4 26B	2	closed	0.172 [0.167, 0.175]	0.189	0.444 [0.370, 0.518]	0.049 [0.037, 0.074]	22 / 93 / 0 / 0	208/252
Gemma 4 26B	2	dashboard	0.175 [0.175, 0.175]	0.175	0.370 [0.370, 0.370]	0.037 [0.037, 0.037]	22 / 104 / 0 / 0	225/252

Schemas: *offered* (four verdicts, evidence-anchored definitions); *removed-only* (offered minus `needs_evidence`); *closed* (`pass/fail` only, evidence-anchored definitions kept); *dashboard* (`pass/fail` only, `pass` redefined as "the obligation is satisfied", warning dropped). The anchor subset has 120 unassessable pairs and no controls; Study 2 rates are pooled over its six firms (126 unassessable pairs). Min answered: the fewest pairs answered in any of the three runs. The verdict columns show every off-schema answer: none occurred on an unassessable pair in any cell.

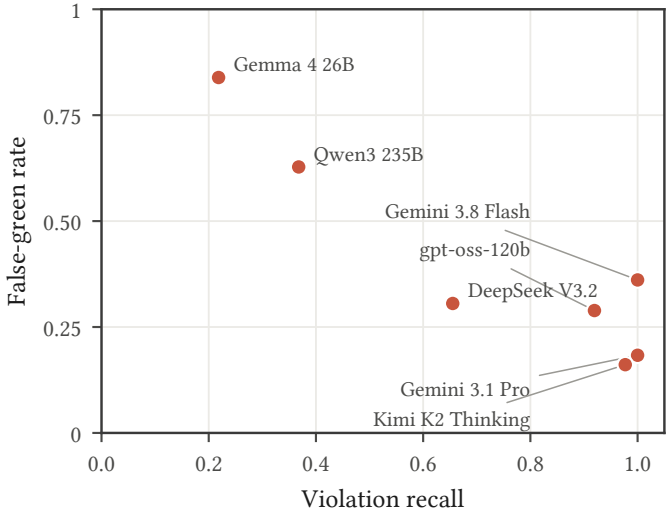


Figure 9. Study 3, proxy-column firms, offered schema: violation recall against false-green rate, one point per model. Higher recall goes with fewer false greens (Spearman −0.55), but no high-recall model is near zero.

Table 14. Paired schema contrasts, repeated-run models.

Model	Study	Change of schema	Unassessable pairs	Change in FGR [95% CI]	Holm p
Gemini 3.1 Pro	1	offered → removed-only	120	-0.008 [-0.04, +0.01]	1.000
Gemini 3.1 Pro	1	offered → closed	120	-0.008 [-0.09, +0.05]	1.000
Gemini 3.1 Pro	1	closed → dashboard	120	+0.081 [+0.02, +0.17]	0.036 **
Gemini 3.1 Pro	1	offered → dashboard	120	+0.072 [-0.04, +0.19]	1.000
Gemini 3.1 Pro	2	offered → removed-only	126	+0.034 [+0.00, +0.11]	1.000
Gemini 3.1 Pro	2	offered → closed	126	+0.040 [+0.00, +0.10]	1.000
Gemini 3.1 Pro	2	closed → dashboard	126	+0.000 [+0.00, +0.00]	1.000
Gemini 3.1 Pro	2	offered → dashboard	126	+0.040 [+0.00, +0.10]	1.000
Gemini 3.8 Flash	1	offered → removed-only	108	+0.077 [-0.01, +0.20]	1.000
Gemini 3.8 Flash	1	offered → closed	115	+0.136 [+0.04, +0.24]	0.019 **
Gemini 3.8 Flash	1	closed → dashboard	103	+0.168 [+0.10, +0.24]	0.012 **
Gemini 3.8 Flash	1	offered → dashboard	108	+0.324 [+0.17, +0.48]	0.012 **
Gemini 3.8 Flash	2	offered → removed-only	120	+0.003 [+0.00, +0.01]	1.000
Gemini 3.8 Flash	2	offered → closed	120	+0.133 [+0.00, +0.33]	0.960
Gemini 3.8 Flash	2	closed → dashboard	126	+0.011 [-0.01, +0.04]	1.000
Gemini 3.8 Flash	2	offered → dashboard	120	+0.142 [+0.00, +0.37]	1.000
Gemma 4 26B	1	offered → removed-only	97	+0.086 [+0.01, +0.20]	0.012 **
Gemma 4 26B	1	offered → closed	102	+0.542 [+0.38, +0.70]	0.012 **
Gemma 4 26B	1	closed → dashboard	102	-0.049 [-0.16, +0.04]	1.000
Gemma 4 26B	1	offered → dashboard	108	+0.509 [+0.34, +0.67]	0.012 **
Gemma 4 26B	2	offered → removed-only	126	+0.000 [+0.00, +0.00]	1.000
Gemma 4 26B	2	offered → closed	109	+0.110 [+0.01, +0.25]	0.320
Gemma 4 26B	2	closed → dashboard	109	+0.000 [+0.00, +0.00]	1.000
Gemma 4 26B	2	offered → dashboard	126	+0.103 [+0.02, +0.24]	0.102

Each unassessable pair's outcome is its share of `pass` over the three runs. Pairs answered in every run of both conditions; obligations resampled (2,000 draws, firm by obligation in Study 2); two-sided p-values Holm-corrected across all 24 contrasts; ** marks $p < 0.05$.

G Interpretive choices of the reference checker

The checker of Section 4.3 records every choice the regulations leave open, with its source, in `labelcheck/INTERPRETATIONS.md`. The choices that matter most are these. Section 1 signed on the first day of employment is timely (8 C.F.R. 274a.2(b)(1)(i)(A); M-274 §3.0). Section 2 is due by the third business day after the first day, with weekends and federal holidays (with their observed dates) excluded; the M-274 does not define "business day". Reverification on the expiry date itself is timely. The required wage is the greater of the prevailing wage and the actual wage, compared per pay period rather than as a year-to-date average (20 C.F.R. 655.731(a), (c)(4)). None of the five proxy columns is treated as evidence. Section 4.3 gives the number of verdicts each boundary choice would change.

Table 15. Study 3, false-green rate by model, firm rendering and schema.

Model	Arm	FGR offered	removed-only	closed	dashboard	Recall, offered	False alarm, offered	Min answered
Gemini 3.1 Pro	plain	0.106	0.189	0.206	0.294	1.000	0.000	504/504
Gemini 3.1 Pro	proxy	0.183	0.206	0.572	0.650	1.000	0.000	504/504
Gemini 3.8 Flash	plain	0.161	0.217	0.350	0.361	1.000	0.000	498/504
Gemini 3.8 Flash	proxy	0.361	0.433	0.639	0.778	1.000	0.000	471/504
Gemma 4 26B	plain	0.339	0.333	0.406	0.411	0.230	0.084	498/504
Gemma 4 26B	proxy	0.839	0.889	0.956	0.894	0.218	0.093	494/504
gpt-oss-120b	plain	0.133	0.239	0.228	0.228	0.954	0.017	474/504
gpt-oss-120b	proxy	0.289	0.228	0.272	0.400	0.920	0.030	468/504
DeepSeek V3.2	plain	0.111	0.211	0.189	0.267	0.529	0.131	504/504
DeepSeek V3.2	proxy	0.306	0.411	0.767	0.772	0.655	0.127	504/504
Qwen3 235B	plain	0.317	0.350	0.367	0.394	0.345	0.194	489/504
Qwen3 235B	proxy	0.628	0.739	0.894	0.917	0.368	0.232	494/504
Kimi K2 Thinking	plain	0.156	0.161	0.261	0.233	0.966	0.000	474/504
Kimi K2 Thinking	proxy	0.161	0.339	0.489	0.583	0.977	0.034	456/504

Twelve fresh firms (504 pairs, 180 unassessable) per arm, one run per cell. *plain*: generator v2; *proxy*: the same firms with the five proxy columns (generator v3). False alarm is any non-`pass` verdict on one of the 237 conforming controls.

Across the checked firms, 70 Section 2 dates fall exactly on the deadline day; counting weekdays without excluding federal holidays would change 6 verdicts, and counting calendar days 55. The 1,176 distinct labels are 1,680 counting the proxy twins of Study 3, whose labels are identical by construction (`docs/labelcheck_results.json`).

Table 16. Study 3, the declared tests.

Estimate	P1 abstention	P2 definition and warning	P3 proxy columns, dashboard	P4 proxy columns, offered
Pooled, 7 models (primary)	+0.259 [+0.20, +0.33] p=0.002 **	+0.084 [+0.04, +0.12] p=0.002 **	+0.427 [+0.29, +0.56] p=0.002 **	+0.217 [+0.15, +0.29] p=0.002 **
Google, 3 models	+0.258 [+0.16, +0.36] p=0.004 **	+0.069 [+0.01, +0.13] p=0.021 **	+0.440 [+0.27, +0.61] p=0.004 **	+0.265 [+0.18, +0.35] p=0.004 **
Other developers, 4 models	+0.260 [+0.18, +0.34] p=0.004 **	+0.095 [+0.04, +0.14] p=0.004 **	+0.417 [+0.29, +0.53] p=0.004 **	+0.181 [+0.12, +0.26] p=0.004 **
Gemini 3.1 Pro	+0.389 [+0.24, +0.55] p=0.014 **	+0.078 [+0.00, +0.17] p=0.432	+0.356 [+0.22, +0.50] p=0.014 **	+0.078 [+0.01, +0.17] p=0.432
Gemini 3.8 Flash	+0.269 [+0.10, +0.46] p=0.014 **	+0.136 [+0.00, +0.27] p=0.432	+0.417 [+0.18, +0.64] p=0.015 **	+0.218 [+0.07, +0.40] p=0.033 **
Gemma 4 26B	+0.117 [+0.00, +0.25] p=0.432	-0.006 [-0.02, +0.00] p=1.000	+0.547 [+0.34, +0.72] p=0.014 **	+0.500 [+0.29, +0.69] p=0.014 **
gpt-oss-120b	-0.020 [-0.12, +0.07] p=1.000	+0.154 [+0.07, +0.23] p=0.014 **	+0.174 [+0.08, +0.25] p=0.014 **	+0.160 [+0.05, +0.30] p=0.015 **
DeepSeek V3.2	+0.461 [+0.28, +0.62] p=0.014 **	+0.006 [-0.06, +0.07] p=1.000	+0.506 [+0.35, +0.66] p=0.014 **	+0.194 [+0.06, +0.36] p=0.015 **
Qwen3 235B	+0.267 [+0.12, +0.40] p=0.014 **	+0.035 [-0.04, +0.11] p=1.000	+0.545 [+0.32, +0.74] p=0.014 **	+0.311 [+0.19, +0.45] p=0.014 **
Kimi K2 Thinking	+0.333 [+0.20, +0.49] p=0.014 **	+0.187 [+0.06, +0.31] p=0.070	+0.444 [+0.25, +0.63] p=0.015 **	+0.058 [-0.03, +0.17] p=1.000

Difference in false-green rate on unassessable pairs answered in both cells, [95% CI] from an obligation-cluster bootstrap (18 firm-by-rule clusters, 2,000 draws), Holm-adjusted p. Holm families as declared: the four pooled tests; the 28 per-model tests; the eight group tests. ** marks adjusted p < 0.05. Pooled estimates weight each model equally.